

Dynamic Disclosure with(out) Timestamps*

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Abstract

We study the role of timestamps in a dynamic disclosure game. At a random date, a sender privately obtains hard evidence about a hidden, evolving binary state and chooses when to disclose it. With timestamps, the unique equilibrium features immediate disclosure of good evidence and a deterministic, timestamp-dependent delay for bad evidence. Without timestamps, if the prior is high relative to sender impatience, good evidence disclosure must be delayed, and we construct an equilibrium in which bad evidence is never disclosed, and good evidence is initially concealed, then stochastically released, and eventually disclosed immediately. By contrast, with an impatient sender, we construct an equilibrium in which bad evidence is disclosed in isolated bursts, while good evidence is disclosed immediately. Timestamps prevent pretending old good evidence is fresh and allow proving bad evidence is old, thereby accelerating disclosure of good evidence and facilitating disclosure of bad evidence.

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1 Introduction

A key feature of many evidentiary documents is whether they carry a timestamp. Evidence may therefore convey not only information about the underlying state, but also information about when that information was generated. Medical test results, bankruptcy filings, audit reports, and legal records such as signed contracts and digital forensic logs often bear verifiable dates. By contrast, many commonly used documents are not timestamped. For example, undated recommendation letters provide an evaluation without indicating when it was written, and sensationalist news media sometimes reuse old photos to support current narratives when original dates are difficult or costly for the public to verify. In addition, evidence of demand may be effectively untimestamped when recorded transaction dates do not reveal when the underlying demand arose, such as in sales “sandbagging,” where ready-to-close deals are deliberately deferred to a later reporting period.

There has been growing attention in the literature to how the timing of disclosure affects beliefs and incentives in dynamic environments. However, existing work typically abstracts from whether the act of disclosure itself reveals the time at which the evidence was generated. It remains an open question whether the presence or absence of timestamps plays a role in strategic disclosure over time.

We address this question by introducing and studying a simple continuous-time sender-receiver model of dynamic disclosure of hard evidence about an evolving state. Specifically, there is a hidden state θ_t that evolves according to Markov switching dynamics between two states, 1 (“good”) and 0 (“bad”). A sender privately obtains one piece of hard evidence at some random, exponentially distributed date τ and can choose to disclose it (or not) at any later time. We consider two distinct environments: *timestamps* and *no timestamps*. In the timestamps environment, upon disclosure, the receiver sees both θ_τ and τ ; in the no-timestamps environment, the receiver only sees θ_τ and must infer τ itself. The sender’s flow payoff is the receiver’s posterior belief that the state is 1 and is discounted at rate r . This simple environment generates rich equilibrium dynamics that differ qualitatively between the timestamps and no-timestamps environments.

In the timestamps environment, we show that there is a unique equilibrium, and good evidence is disclosed immediately while bad evidence is disclosed after a deterministic, timestamp-dependent delay. To understand the intuition, consider first the incentive to disclose good evidence. If the sender possesses good evidence $\theta_\tau = 1$ at τ and discloses it immediately, the receiver’s belief jumps to 1 and decays according to the hidden Markov switching dynamics thereafter. If instead the sender were to delay until some time $t > \tau$, the receiver would retrospectively update about θ_τ , so her belief path after t would be exactly

the same as under immediate disclosure. In equilibrium, the only effect of delay is to lower the receiver’s belief during the interval $[\tau, t)$, while evidence is being withheld.¹ Timestamps prevent the sender from pretending good evidence is fresh, and thus the sender cannot benefit from delaying disclosure of good evidence.

In addition to proving that good evidence is fresh, timestamps allow the sender to prove that old bad evidence is in fact old. Therefore, when the sender discloses old bad evidence, beliefs update retrospectively downward to 0, but drift up according to the hidden Markov switching after that. Furthermore, since evidence only arrives once, once bad evidence has been disclosed, the receiver no longer expects good evidence to be disclosed. This shuts down a “no news is bad news” effect which raises the trajectory of future beliefs. For each bad evidence arrival date, there is a unique date at which the receiver’s belief after disclosure of that evidence crosses the no-disclosure belief path from below. This is the optimal time to disclose bad evidence. Although the full no-disclosure path depends on the disclosure strategy for each piece of evidence, which in turn depends on the no-disclosure path, we introduce an auxiliary belief that allows us to reduce the problem to a set of one-dimensional fixed point problems.

A property of the unique equilibrium under timestamps is that all evidence is disclosed at the first instant it is myopically optimal to do so; hence, the equilibrium is independent of the sender’s discount rate. Additionally, we show that the Minimum Principle (Acharya, DeMarzo and Kremer, 2011; Guttman, Kremer and Skrzypacz, 2014) applies in this timestamped setting: at all times, the receiver’s equilibrium belief conditional on no disclosure is the most pessimistic belief consistent with Bayes’ rule among all possible disclosure strategies. Neither of these properties holds outside of the timestamps environment.

In the no-timestamps environment, we first show that if the prior belief is sufficiently high relative to the sender’s discount rate, equilibrium must feature delayed disclosure of some good evidence. If the receiver conjectured immediate disclosure of good evidence at all times, a sender with good evidence could disclose it after a brief delay and convince the receiver that the evidence is fresh. The sender would obtain a lower flow payoff during the delay, but a higher payoff at all future times. When the prior belief is above a cutoff, this trade-off favors delay.²

To cleanly characterize delayed disclosure of good evidence, we specialize to the case

¹For some non-equilibrium conjectures, delay could temporarily raise the receiver’s belief; our formal argument rules these out.

²In standard disclosure models, equilibria often feature cutoff strategies, with highest types disclosing immediately. The delayed disclosure of good evidence here does not contradict this theme, because the notion of “type” applies not only to whether evidence is good or bad but also its arrival date, and from this perspective, there is no highest type — there are always future arrival dates for the sender to mimic.

where the bad state is absorbing, which is sufficient for ensuring bad evidence is never disclosed.³ When the prior is high relative to sender impatience, we construct an equilibrium consisting of three distinct phases, which we label *stockpiling*, *purging*, and *transparency*. In the stockpiling phase, the sender does not disclose good evidence, creating a stockpile of (possible arrival dates of) undisclosed good evidence from the receiver’s perspective. At some date, the sender would prefer to disclose good evidence. However, equilibrium cannot have a bang-bang disclosure strategy with respect to good evidence, with an abrupt transition to a phase of immediate disclosure, because at that transition, the receiver would be expecting all old good evidence to be disclosed, and therefore would not revise her belief to 1. The sender could simply wait an instant longer and separate from these older evidence types.

This tension — where a conjecture of complete disclosure incentivizes further delay — is naturally resolved with mixing in the purging phase. During this phase, the sender stochastically discloses old evidence at an intensity that depends on calendar time but not on the evidence’s arrival date. Over time, the disclosure intensity increases, which causes the stockpile to deplete, shifting the distribution of evidence to more recent dates. This means that disclosure is eventually interpreted more favorably, which sustains the sender’s indifference to delay in the first place. By the end of this phase, the disclosure intensity explodes and the stockpile fully (but continuously) empties. At that point, the receiver’s belief is sufficiently low that the sender is willing to disclose any good evidence immediately.

The sender’s incentive to delay in the purging phase depends not only on the receiver’s belief about the current state, but also on the receiver’s beliefs about the sender’s evidence, since the latter belief determines the receiver’s belief after disclosure and the evolution of the receiver’s belief conditional on no disclosure. Technically, this presents a challenge in that the receiver’s current belief about the state is not the only state variable. Moreover, the receiver must form a belief distribution at each instant about the continuum of past evidence arrival dates. However, with timestamp-independent disclosure rates, we show that these beliefs can be summarized by a few stock variables that track the joint probability that the sender has good or bad evidence and the state is good, conditional on no disclosure. We describe the rest of the construction in detail in Section 4.2.

Finally, we analyze disclosure of bad evidence, focusing on an impatient sender, which sustains immediate disclosure of good evidence and streamlines comparison to the timestamps environment. Using a stationary prior, we construct an equilibrium in which good evidence is disclosed immediately, while bad evidence is disclosed with delay in isolated bursts. Intuitively, a transparency phase cannot exist for bad evidence, since all disclosures

³We show that the construction goes through when the bad state is not absorbing but is sufficiently persistent.

would be interpreted as fresh. The delays between bursts ensure that disclosed evidence has sufficiently aged on average that disclosure is optimal.

Our analysis has implications for receiver welfare and policy when we microfound the sender’s flow payoff as the result of a population of receivers taking actions attempting to match the state to minimize a quadratic loss. Timestamps not only convey information directly, strictly benefiting receivers at times when multiple evidence arrival-dates would pool without timestamps, but also change strategic disclosure incentives. Without timestamps, there can be delayed disclosure of good evidence, so institutions that require timestamps can help the receiver get information sooner. Moreover, timestamps can facilitate the disclosure of bad evidence (when the bad state is not absorbing) by eliminating equilibria in which no bad evidence is ever disclosed and ensuring that all bad evidence is eventually disclosed.

A brief roadmap is as follows. The remainder of this section discusses related literature. Section 2 introduces the model. Sections 3 and 4 analyze the timestamps and no-timestamps environments, respectively, and present our main results. Section 5 concludes with a discussion of receiver welfare, modeling assumptions, and robustness.

Related literature

Our paper contributes to a large literature on strategic disclosure of hard information by a sender without commitment, pioneered by [Grossman and Hart \(1980\)](#), [Grossman \(1981\)](#), and [Milgrom \(1981\)](#). While the canonical static models predict full unraveling, with essentially all sender types disclosing to separate themselves from lower types, [Dye \(1985\)](#) showed that when there is uncertainty about whether the sender has evidence to disclose, unraveling need not be complete, as low types with evidence can refrain from disclosing it to mimic high types that do not have evidence. A long literature on mostly static models has explored various alternative explanations for nondisclosure.

More recent literature has studied disclosure in dynamic settings. In [Acharya et al. \(2011\)](#), a sender chooses when to disclose information about a fixed state of the world when there is exogenous public news about that state. The authors show that exogenous news creates a real options problem for the sender: by waiting to disclose, the sender retains the option to not disclose in case very favorable news arrives.

In [Guttman et al. \(2014\)](#), the sender can obtain multiple pieces of evidence sequentially across two stages, but the state (firm value) is constant and independent of the evidence arrival times. The authors show that late disclosures are interpreted more favorably. [Antić and Pei \(2026\)](#) study selective disclosure in an overlapping generations model with multiple, noisy pieces of evidence. In their model, evidence does not carry a timestamp. Although

the state is fixed, the absence of timestamps is relevant in that disclosed signals convey information about the history of evidence arrival and what signals could have been concealed. Our focus is different from these papers; evidence arrival time and the current state are intimately linked due to hidden evolution of the state. This allows us to study optimal disclosure timing even when there is only one piece of evidence.

Kremer, Schreiber and Skrzypacz (2024) study disclosure of evidence about an evolving state. In their model, evidence is effectively timestamped, but the sender must either disclose it immediately or never; delayed disclosure is not permitted, consistent with Securities and Exchange Commission rules. In our model, delayed disclosure is permitted and is central to the analysis: with timestamps, bad evidence is optimally disclosed only after a delay, while without timestamps, good evidence may be delayed in order to appear fresh. Thus, relative to Kremer et al. (2024), our paper isolates how the ability to delay disclosure interacts with the presence or absence of timestamps. A further difference relates to the threshold for disclosing evidence. In Kremer et al. (2024), this threshold lies below the current mean belief of the receiver. In our model, with timestamps, bad evidence is disclosed only once it is old enough that disclosure doesn't lower the receiver's belief about the current state. Furthermore, without timestamps, good evidence is sometimes withheld even though disclosure would immediately raise the receiver's belief.

Other papers study dynamic persuasion with commitment; e.g., Ely (2017) and Ely, Georgiadis and Rayo (2025). In the latter paper, feedback is given to the sender (receiver) according to a bang-bang rule: first silence, then transparency. Although the results are not directly comparable due to fundamental model differences, it is notable that in our no-timestamps environment, the bang-bang policy cannot occur in equilibrium, and instead there is a purging phase between our stockpiling (silence) and transparency phases.

2 Model

The game is played between a sender and a receiver in continuous time over an infinite horizon. There is a hidden, binary underlying state $\theta_t \in \{0, 1\}$ that evolves according to Markov switching; when the state is i , it switches to $j \neq i$ at constant exponential rate λ_i . We assume $\lambda_1 > 0$ and $\lambda_0 \geq 0$. Neither the sender nor the receiver observes the realization of the state process, and they share a common prior over the initial state $p_0 = \Pr(\theta_0 = 1) \in (0, 1)$.

At some random date τ , the sender privately obtains hard evidence of the current state θ_τ . The receiver does not directly observe τ or θ_τ . We assume that τ is distributed exponentially with parameter μ and is independent of the state process $(\theta_t)_{t \geq 0}$. The sender can receive evidence at most once. The sender then chooses to disclose this evidence at any date $s \geq \tau$

or never disclose it, which we denote by $s = \infty$. We consider two environments: *timestamps* and *no timestamps*. In the timestamps environment, when evidence is disclosed, the receiver observes both the realization θ_τ and the timestamp τ . In the no-timestamps environment, the receiver only observes the realization θ_τ (and must infer τ).

A strategy for the sender consists of a collection of cumulative distribution functions $H(\cdot; \theta_\tau, \tau, t)$ on $[t, \infty]$, for $\theta_\tau \in \{0, 1\}$ and $0 \leq \tau \leq t$, where $H(s; \theta_\tau, \tau, t)$ specifies the probability that a sender who obtained evidence θ_τ at time τ and has not disclosed it before time t plans to disclose by time s . Strategies must satisfy the usual internal consistency requirement. We say that a strategy is *age-independent* if for all τ, τ', t with $0 \leq \tau, \tau' \leq t$, and $x \in \{0, 1\}$ $H(\cdot; x, \tau, t) = H(\cdot; x, \tau', t)$. In other words, the continuation strategy is independent of the evidence's age conditional on the current time. Much of our analysis in the no-timestamps environment focuses on age-independent equilibria.

A belief system for the receiver specifies at each time t , given the public history, a joint distribution over the current state θ_t and the sender's evidence history E_t , where $E_t = \emptyset$ denotes no evidence has arrived and $E_t = (\tau, x) \in [0, t] \times \{0, 1\}$ if evidence arrived at time τ and has type $x = \theta_\tau$. We denote by p_t the receiver's posterior belief at time t that $\theta_t = 1$; the process $\pi = (\pi_t)_{t \geq 0}$ is $[0, 1]$ -valued and càdlàg. The sender's objective is to maximize

$$\mathbb{E} \left[\int_0^\infty e^{-rt} \pi_t dt \right]$$

where $r > 0$ is the discount rate. Note that our receiver is passive, simply performing Bayesian updating; in Section 5, we provide a microfoundation and discuss receiver welfare.

A perfect Bayesian equilibrium (henceforth, equilibrium) is a strategy and belief system such that the sender's continuation strategy $H(\cdot; x, \tau, t)$ is sequentially rational for all (x, τ, t) with $t \geq \tau$, and the belief process is consistent with Bayes' rule whenever possible. Since evidence arrives stochastically, nondisclosure is always on path, and Bayes' rule applies. Disclosure can be off-path. In the timestamps case, the receiver's belief after off-path disclosure is already determined by the timestamp.⁴ In the no-timestamps case, there is flexibility in off-path beliefs.

Belief evolution and focal cases. Let $p^* := \lambda_0/(\lambda_0 + \lambda_1)$ denote the long-run steady-state belief. Given the continuous-time Markov chain, two belief processes will be useful throughout. First,

$$\phi_t := p^* + (p_0 - p^*)e^{-(\lambda_0 + \lambda_1)t}$$

⁴Recall that the sender does not observe the state evolution, and therefore the receiver's off-path beliefs are pinned down by her beliefs about the evidence arrival date. This is true with and without timestamps.

is the ex ante probability that $\theta_t = 1$, which converges monotonically to p^* from above (below) when $p_0 > p^*$ ($p_0 < p^*$). Second, adopting the convention that G (“good”) refers to state 1 and B (“bad”) to state 0, define $q_t^{G,s} := \Pr(\theta_t = 1 \mid \theta_s = 1)$ and $q_t^{B,s} := \Pr(\theta_t = 1 \mid \theta_s = 0)$ as the probabilities that $\theta_t = 1$ conditional on the state at $s \leq t$ being good or bad, respectively. Then for all $t \geq s \geq 0$ and $i \in \{G, B\}$, beliefs evolve according to

$$\dot{q}_t^{i,s} = \lambda_0(1 - q_t^{i,s}) - \lambda_1 q_t^{i,s}$$

with initial conditions $q_s^{G,s} = 1$ and $q_s^{B,s} = 0$. These admit closed-form solutions

$$q_t^{G,s} = \frac{\lambda_0}{\lambda_0 + \lambda_1} + \frac{\lambda_1}{\lambda_0 + \lambda_1} e^{-(\lambda_0 + \lambda_1)(t-s)} \quad \text{and} \quad q_t^{B,s} = \frac{\lambda_0}{\lambda_0 + \lambda_1} - \frac{\lambda_0}{\lambda_0 + \lambda_1} e^{-(\lambda_0 + \lambda_1)(t-s)}. \quad (1)$$

As $t \rightarrow \infty$, $q_t^{G,s}$ converges to p^* from above and $q_t^{B,s}$ converges to p^* from below.

Two particular cases are useful for our analysis. First, when $\lambda_0 = 0$, the bad state is absorbing. This corresponds to settings where, for example, the state is the quality or value of a technology that eventually experiences a permanent breakdown. An implication is that bad evidence disclosure is never optimal, with or without timestamps. This case allows us to isolate the role of timestamps in strategic disclosure of good evidence. Second, when $p_0 = p^*$, which we refer to as the stationary prior case, the long-run unconditional belief is constant: $\phi_t = p^*$ at all times. This shuts down the effect of the prior and allows us to isolate dynamics arising from stochastic evidence arrival and strategic disclosure alone.

3 Timestamps

3.1 Equilibrium characterization

In this section, we show that there exists a unique equilibrium under timestamps: good evidence is disclosed immediately, while bad evidence is disclosed after a deterministic, timestamp-dependent delay. This result is formalized in Theorem 1.

For intuition, suppose the sender obtains good evidence at time s . The sender could choose to disclose it immediately and raise the receiver’s belief immediately to 1, after which it would decay toward the steady state due to the hidden Markov switching. Suppose instead the sender chooses to disclose at time $t > s$. Importantly, since the evidence is timestamped, the receiver learns that $\theta_s = 1$, and the resulting belief is the same for all $u \geq t$ as it would have been after disclosure at time s . Moreover, the sender would obtain a lower flow payoff on $[s, t)$: under the conjectured equilibrium strategy, no news is weakly bad news, so the receiver’s nondisclosure belief would be weakly lower than the belief given disclosure at time

s . A subtlety is that immediate disclosure of good evidence is not a best response for *all* conjectures of the sender's strategy: there exist conjectures for which the ranking of beliefs on $[s, t)$ need not hold. However, these conjectures would involve disclosure of bad evidence at a time when it would be suboptimal; the proof of Theorem 1 rules them out.

To understand why bad evidence must be disclosed eventually when the bad state is not absorbing, consider an equilibrium with no disclosure of bad evidence but immediate disclosure of good evidence. Then nondisclosure reveals that either the sender has no evidence or has bad evidence. Over time, the probability on no evidence vanishes, and a sender with old bad evidence eventually finds it optimal to disclose that evidence to separate himself from types with more recent bad evidence. Put another way, conditional on no disclosure, “no news is bad news” relative to the belief dynamics based on Markov switching alone. In contrast, once bad evidence is disclosed, the receiver understands that no good evidence can be disclosed, shutting down this effect. Hence, the belief after disclosing bad evidence eventually crosses the no-disclosure belief from below. The sender need not trade-off any short-run cost of disclosure; he can simply wait until the time that the belief paths cross.

This logic generalizes: as long as the bad state is not absorbing, for each timestamp s , the sender eventually finds it worthwhile to disclose bad evidence at the time t when

$$q_t^{B,s} = p_t, \quad (2)$$

where the right-hand side of (2) is the equilibrium no-disclosure belief. As we show, for all s , $q_t^{B,s}$ crosses p_t from below and only once, so disclosing at the crossing time is optimal independent of the sender's discount rate.

Since the full path $(p_t)_{t \geq 0}$ is an endogenous object that depends on the exact disclosure time for each piece of evidence, the equilibrium characterization a priori requires an infinite-dimensional fixed point. However, by constructing a useful auxiliary belief, the problem reduces to a continuum of independent one-dimensional equations. To that end, we establish and exploit the following strict monotonicity property: at the instant each piece of bad evidence is disclosed, all earlier bad evidence will have been already disclosed, and no later bad evidence will have been disclosed. For each fixed s and all $t \geq s$, define

$$p_{t,s}^\dagger := \frac{e^{-\mu t} \phi_t + \int_s^t \mu e^{-\mu \tau} (1 - \phi_\tau) q_t^{B,\tau} d\tau}{e^{-\mu t} + \int_s^t \mu e^{-\mu \tau} (1 - \phi_\tau) d\tau} \quad (3)$$

as the belief at time t , conditional on no disclosure, under a conjecture that all good evidence is disclosed immediately, and bad evidence with $\tau \in [0, s]$ would be disclosed by time t , and no bad evidence with timestamp $\tau > s$ would be disclosed by time t .

For each fixed s , $(p_{t,s}^\dagger)_{t \geq s}$ and $(p_t)_{t \geq s}$ need not coincide since they are based on different conjectures. However, the key property is that whenever the optimality condition (2) holds, they coincide. Hence, instead of fully characterizing $(p_t)_{t \geq 0}$ before solving (2) for each s , we can solve the equations (in t)

$$q_t^{B,s} = p_{t,s}^\dagger. \quad (4)$$

The proof of our result establishes a unique solution (4) for all $s \geq 0$, as long as $\lambda_0 > 0$.

Theorem 1. *There exists a unique equilibrium. If the sender ever possesses good evidence (on or off path), he discloses it immediately. If $\lambda_0 = 0$, bad evidence is never disclosed. For all $\lambda_0 > 0$, if the sender obtains bad evidence at time s , he discloses it at time $s + a^*(s)$, where $a^*(s) \in (0, \infty)$ for all $s \geq 0$ is the unique positive solution to*

$$q_{s+a^*(s)}^{B,s} = p_{s+a^*(s),s}^\dagger.$$

If the sender still possesses this evidence after $s + a^(s)$ (off path), he discloses it immediately.*

The equilibrium belief p_t is therefore piecewise. Prior to $t < a^*(0)$, no bad evidence is disclosed, hence $p_t = p_{t,0}^\dagger$. For $t \geq a^*(0)$, bad evidence with earlier timestamps has already been disclosed, whereas bad evidence with later timestamps is still being withheld. Hence, $p_t = p_{s+a^*(s),s}^\dagger$ where $t = s + a^*(s)$.

Overall, timestamps prevent the sender from pretending old good evidence is fresh, and allow the sender to prove old bad evidence is old. The equilibrium no-disclosure belief p_t converges to a level strictly below the steady state of the Markov chain, a topic we explore in the next subsection.

Figure 1 illustrates the equilibrium belief dynamics under a high prior (left panel) and a low prior (right panel). The black curve is the receiver's belief conditional on no disclosure, p_t . If good evidence arrives at time s_1 , the sender discloses it immediately and the receiver's belief jumps to 1 and then follows the blue path. If bad evidence arrives at time s , where $s = s_1$ or s_2 , the sender waits until the red post-disclosure belief path first reaches the black no-disclosure path. This crossing date is $s + a^*(s)$.

3.2 Equilibrium properties

We now examine how the timing of bad evidence disclosure changes with its timestamp and its implications for the equilibrium beliefs.

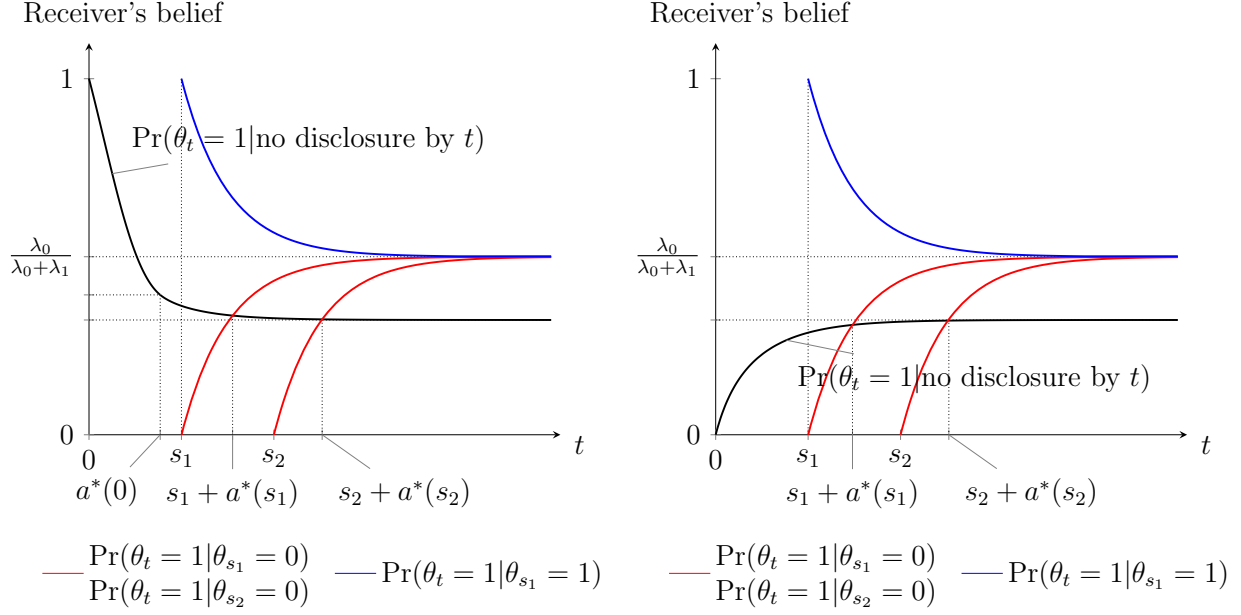


Figure 1: Equilibrium belief evolution for limiting case priors $p_0 = 1$ (left panel) and $p_0 = 0$ (right panel), and parameters $\lambda_0 = 1, \lambda_1 = 1, \mu = 0.5$. The equilibrium is independent of r .

Proposition 1. Assume $\lambda_0 > 0$. For all $s \geq 0$, (i) $s + a^*(s)$ is increasing in s ; (ii) $a^*(s)$ is monotone in s : it is strictly increasing in s if $p_0 < p^*$, strictly decreasing if $p_0 > p^*$, and constant if $p_0 = p^*$; and (iii) $\lim_{s \rightarrow \infty} a^*(s) = \alpha^* \in (0, \infty)$.

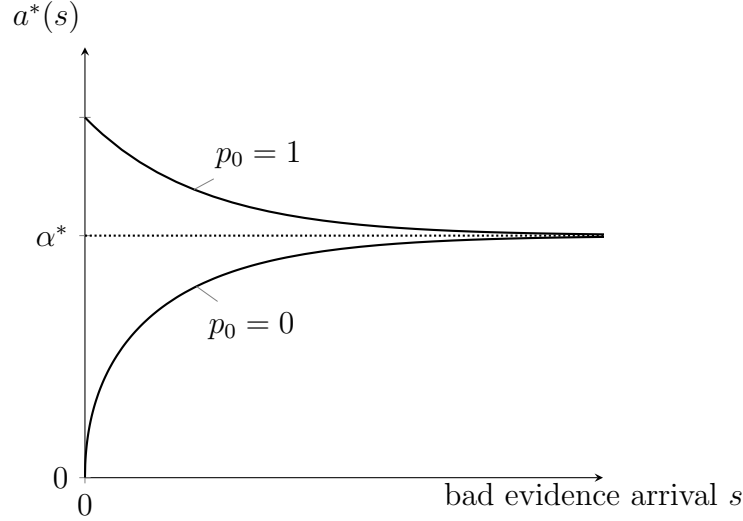


Figure 2: Optimal delay $a^*(s)$ as a function of s for priors $p_0 = 1$ and $p_0 = 0$, and parameters $\lambda_0 = 1, \lambda_1 = 1, \mu = 0.5$.

The fact that optimal disclosure time $s + a^*(s)$ is increasing in the timestamp s is due to the crossing condition (2) and the fact that older evidence is less damaging than new; hence,

older evidence becomes worth disclosing first.

To understand why $a^*(s)$ is monotone, fix a delay a and compare the disclosure condition at different calendar times. If bad evidence timestamped at s is disclosed at $s + a$, its disclosure value depends only on the delay a and not the calendar time because the Markov process is time-homogeneous. Calendar time therefore affects the disclosure condition only through the value of nondisclosure. For the marginal bad-evidence type, no disclosure pools no evidence with bad evidence that arrived within the last a units of time. When $p_0 < p^*$, the unconditional belief is increasing, so no evidence is better news at later dates, and evidence arriving during the later interval is less likely to be bad. No disclosure is therefore more valuable at later dates, so bad evidence must age longer for the belief upon disclosing it to catch up, so $a^*(s)$ is increasing. When $p_0 > p^*$, the reverse holds.

The behavior of $a^*(s)$ in turn determines the behavior of the equilibrium no-disclosure belief once bad-evidence disclosure begins. As Figure 1 illustrates, for $t \geq a^*(0)$, the no-disclosure belief moves in the same direction as $a^*(s)$. At the optimal disclosure date, the no-disclosure belief equals the posterior induced by the marginal bad evidence. When $p_0 < p^*$, $a^*(s)$ is increasing, so the marginal bad evidence disclosed at later dates is older and therefore induces a higher post-disclosure belief. When $p_0 > p^*$, $a^*(s)$ is decreasing, the reverse holds.⁵

The convergence of $a^*(s)$ also determines the long-run no-disclosure belief. As $a^*(s)$ converges to α^* , the age of the marginal bad evidence converges to α^* , so the no-disclosure belief converges to the belief induced by bad evidence of that age. As alluded to in the previous subsection, this limit is strictly below the unconditional steady-state belief p^* , which means no disclosure remains informative even in the long run. At large t , there is still a positive probability that the sender received bad evidence within the last α^* units of time and has not yet disclosed it. The following corollary formalizes these results.

Corollary 1. *At each $t \geq a^*(0)$, the equilibrium no-disclosure belief p_t is strictly increasing in t if $p_0 < p^*$, strictly decreasing in t if $p_0 > p^*$, and constant in t if $p_0 = p^*$. Moreover, $\lim_{t \rightarrow \infty} p_t = p^* (1 - e^{-(\lambda_0 + \lambda_1)\alpha^*}) < p^*$.*

We next show that the receiver’s belief conditional on nondisclosure is minimized in the equilibrium among all possible disclosure strategies, with or without timestamps. This result extends the *Minimum Principle* from earlier work of Acharya et al. (2011) and Guttman et al. (2014).⁶ Intuitively, for any fixed time t , the delayed disclosure of bad evidence induces a “cutoff type” in the form of the timestamp of bad evidence such that the sender is indifferent between disclosing at time t or not. All bad evidence with timestamp before this cutoff, along

⁵The claim that the equilibrium no-disclosure belief p_t is monotone applies only after bad-evidence disclosure begins, at $t \geq a^*(0)$; before $a^*(0)$, it may not be monotone.

⁶A related “suspicious beliefs” principle holds in Kremer et al. (2024).

with all good evidence of any timestamp, is disclosed in equilibrium by time t and would lead to a higher belief at t than p_t ; conversely, all bad evidence with timestamp after the cutoff would lead to a lower belief at t than p_t . Hence, any alternative strategy would either involve nondisclosure of more favorable evidence or disclosure of less favorable evidence (or both). Both kinds of deviations would improve the nondisclosure pool in the sense of raising the average time- t belief conditional on nondisclosure.

Proposition 2 (The Minimum Principle). *At each $t \geq 0$, the equilibrium no-disclosure belief p_t is the minimum among all disclosure strategies.*

As we shall see, the Minimum Principle does not generally hold without timestamps.

4 No timestamps

Without timestamps, the sender can attempt to manipulate the receiver's belief about the age of evidence by strategically timing its disclosure. Since this manifests differently for senders with good versus bad evidence, we divide our analysis into two parts.

4.1 Good evidence: delay and gradual purging

We begin with a simple but general result that says that when the prior is sufficiently high relative to the sender's impatience, there *must* be delayed disclosure of good evidence, in contrast to the timestamps environment.

Proposition 3. *If $p_0 > \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}$, then there is no equilibrium with immediate disclosure of good evidence at all times.*

To understand Proposition 3 and the cutoff belief, suppose, to the contrary, that there is an equilibrium in which good evidence is disclosed immediately and consider a sender who obtains good evidence at time 0. If the sender deviates and discloses at some small time $\varepsilon > 0$, then on $[0, \varepsilon)$, the posterior belief is lower than had the sender disclosed at time 0. Since beliefs are right-continuous, to a first-order approximation, the loss from this delay is $(1 - p_0)\varepsilon$. However, when the receiver sees the disclosure at time ε , she updates to 1 assuming that the evidence is fresh. Compared to having disclosed at time 0, the belief at time ε is higher by approximately $\lambda_1\varepsilon$, where $-\lambda_1$ is the slope of the post-disclosure belief at time 0. Accounting for discounting and mean reversion after the disclosure, the net present value of this difference is $\lambda_1\varepsilon \frac{1}{\lambda_0 + \lambda_1 + r}$. Put together, delay is optimal if $(1 - p_0)\varepsilon < \lambda_1\varepsilon \frac{1}{\lambda_0 + \lambda_1 + r}$, which rearranges to the condition in Proposition 3. Intuitively, the trade-off favors delay when the prior is high.

Proposition 3 also implies that, when the prior is sufficiently high, the Minimum Principle fails without timestamps. Delay means that the nondisclosure pool now contains some good evidence types, raising the nondisclosure belief.

Although Proposition 3 says that equilibrium must sometimes feature delayed disclosure of good evidence, it does not say how disclosure occurs. Lemma 1 establishes that good evidence cannot follow a “bang-bang” pattern, with delay followed by immediate disclosure. The reason is intuitive. Suppose the receiver conjectured full disclosure beginning at time $t_0 > 0$. Then given disclosure at t_0 , the receiver’s belief would jump to a value less than 1, based on her belief distribution over possible evidence ages. By delaying until $t_0 + \epsilon > 0$, the sender can induce a posterior belief arbitrarily close to 1, since the receiver assumes the evidence arrived after T . This deviation is profitable as it yields a discrete gain at arbitrarily small cost. In the equilibrium we construct, this dilemma is resolved through mixing.

Lemma 1. *There is no equilibrium in which good evidence disclosure is delayed until some time $t_0 > 0$ and disclosed immediately at all $t \geq t_0$.*

To obtain a clean characterization of good evidence disclosure, suppose now that $\lambda_0 = 0$; that is, the bad state is absorbing. (We return to $\lambda_0 > 0$ later in the section.) Then disclosure of bad evidence is strictly dominated because it guarantees a flow payoff of 0 forever after, isolating the role of good evidence. The following class of equilibria is central to our analysis.

Definition 1. *A three-phase equilibrium is characterized by two deterministic times $\underline{T}, \bar{T}$ with $0 \leq \underline{T} \leq \bar{T} < \infty$ such that*

- Stockpiling phase: *On $[0, \underline{T})$, there is no disclosure of good evidence. Belief updating immediately after off-path disclosure of good evidence is neutral: it is the probability that the state is good conditional on the sender possessing good evidence.*
- Purging phase: *On $[\underline{T}, \bar{T})$, the disclosure of good evidence is atomless and has hazard rate $\kappa_t > 0$ on $(\underline{T}, \bar{T})$ independent of its timestamp, where κ is continuous.*
- Transparency phase: *On $[\bar{T}, \infty)$, the sender discloses good evidence immediately.*

In particular, our stockpiling phase does not rely on belief threats to punish deviations; neutral beliefs are consistent with a post-disclosure posterior belief distribution over arrival dates identical to the pre-disclosure distribution.⁷

We now present the main result of this section, which establishes a unique equilibrium within the three-phase class. Define $\bar{p} := \frac{r + \frac{\lambda_1}{2}}{r + \lambda_1}$ and $\underline{p} := \frac{r}{r + \lambda_1}$.

⁷Neutral beliefs are the limit of strategies with age-independent trembles; that is, for each evidence arrival date $\tau \in [0, \infty)$, the sender mixes on $[\tau, \underline{T})$ by disclosing at constant hazard rate ϵ .

Theorem 2. Assume $\lambda_0 = 0$. For all p_0, λ_1, μ and r , there exists a unique three-phase equilibrium. Moreover,

- If $p_0 > \bar{p}$, all three phases are nonempty.
- If $p_0 \in (\underline{p}, \bar{p}]$, then only the purging and transparency phases are nonempty.
- If $p_0 \leq \underline{p}$, then only the transparency phase is nonempty.

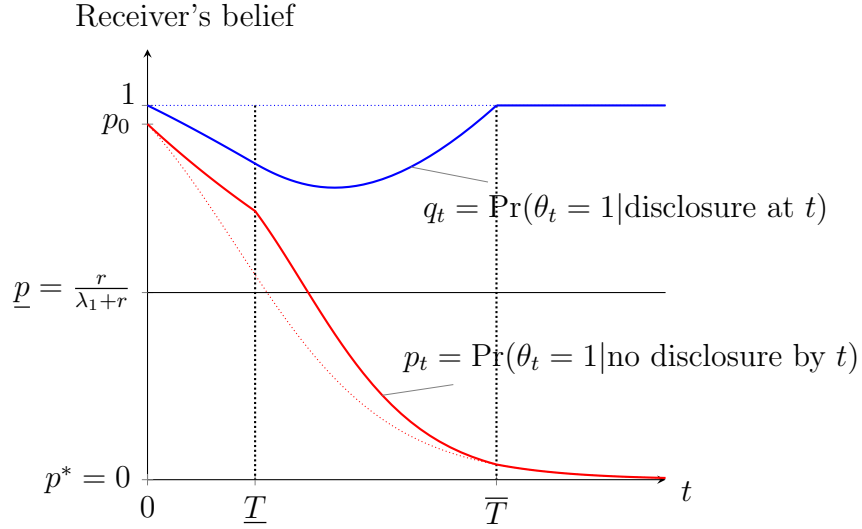


Figure 3: Belief paths in the three-phase equilibrium for $p_0 > \bar{p}$, with vertical dashed lines at $\underline{T}$ and $\bar{T}$. Parameter values: $\lambda_1 = r = 0.5$, $\mu = 2$, $p_0 = 0.95$.

Figure 3 illustrates the equilibrium beliefs when $p_0 > \bar{p}$; p_t denotes the receiver's belief about θ_t conditional on no disclosure, and q_t denotes the receiver's posterior immediately after disclosure of good evidence.⁸ For comparison, the dotted red and blue curves show respectively the receiver's beliefs conditional on no disclosure and disclosure in the unique equilibrium with timestamps. As implied by the Minimum Principle, the no-disclosure belief with timestamps is weakly lower at every t than its counterpart in this three-phase equilibrium without timestamps.

In the stockpiling phase, delay leads to an accumulation or stock of undisclosed evidence from the receiver's perspective. The receiver's belief p conditional on no disclosure falls due to Markov switching, but beyond that, “no news is no news” as the receiver anticipates delay. Meanwhile, q falls due to the aging of evidence, but this fall is dampened by the arrival of new evidence.

⁸Note that while the belief jumps to q_t at time t , after that, it evolves according to Markov switching; i.e., the receiver's belief does not continue to be given by q at all future times after the disclosure.

The purging phase begins at the time $\underline{T}$ when, if the receiver had counterfactually continued to conjecture delay, the sender would strictly prefer to disclose.⁹ At the beginning of the purging phase, p has a downward kink as the rate of disclosure is bounded away from 0. During this phase, q initially continues to fall as the rate of disclosure is small and does not offset the aging of the stock of undisclosed evidence. However, later in the purging phase, p falls sufficiently low that indifference can only be sustained with an increasing q ; this requires κ to increase. Moreover, for q to approach its maximum value 1, it must be that the composition of the pool of undisclosed evidence shifts toward fresh evidence; this requires κ to tend to infinity as $t \rightarrow \bar{T}$. When the transparency phase begins, it is no longer possible to maintain indifference, and the sender discloses good evidence immediately as it arrives.

In Figure 3, it is noteworthy that the belief crosses $\underline{p}$ before the end of the purging phase. In other words, the sender continues to (stochastically) delay even though the receiver's belief p_t conditional on non-disclosure is sufficiently low that, had the game started at the same belief, there would be immediate disclosure at all times. The reason is that the incentive to delay depends not only on the current belief p_t but also the post-disclosure value q_t and its evolution; the presence of a stockpile of undisclosed evidence means that immediate disclosure of good evidence does not convince the receiver that the current state is 1, and delay can be weakly optimal. Thus, the variable q is a critical additional factor in the sender's best response problem.

4.2 Three-phase equilibrium construction

Recall the definition of q_t as the receiver's belief immediately after disclosure of good evidence. Clearly, $q_0 = 1$. If $t \in (0, \underline{T})$, disclosure at time t is off-path, and neutral belief updating means that the receiver's posterior belief over the timestamp is equal to the prior belief about the timestamp given that the evidence was good and there has been no disclosure before time t . Rather than maintain a belief over the whole interval of past times, it is convenient to introduce "stock" variables

$$G_{i,t} = \Pr(\tau \leq t \text{ and } \theta_\tau = 1 \text{ and } \theta_t = i | \text{no disclosure up to } t) \quad \text{for } i \in \{0, 1\},$$

representing the probability that the sender possesses good evidence *and* the state is currently i , conditional on no disclosure yet.

⁹Recall that our three-phase equilibrium definition uses neutral off-path beliefs. Although an incentive to delay past $\underline{T}$ could be sustained for some time by more pessimistic off-path beliefs, the equilibrium conditions and atomless mixing immediately after $\underline{T}$ rule out longer delay; in other words, $\underline{T}$ is pinned down regardless of off-path beliefs, unless one goes beyond this class of equilibria to allow for atoms at the start of purging.

It follows that for $t \in (0, \underline{T})$,

$$q_t = \frac{G_{1,t}}{G_{1,t} + G_{0,t}}.$$

The same characterization also applies during the purging phase, since the disclosure rate of good evidence is independent of its timestamp.

In either the stockpiling or purging phase, (G_1, G_0, p) evolve according to

$$\dot{G}_{1,t} = -G_{1,t}\lambda_1 + (p_t - G_{1,t})\mu - G_{1,t}\kappa_t(1 - G_{1,t} - G_{0,t}) \quad (5)$$

$$\dot{G}_{0,t} = G_{1,t}\lambda_1 - G_{0,t}\kappa_t(1 - G_{1,t} - G_{0,t}) \quad (6)$$

$$\dot{p}_t = -\lambda_1 p_t - G_{1,t}\kappa_t + p_t(G_{0,t} + G_{1,t})\kappa_t, \quad (7)$$

with initial values $(0, 0, p_0)$. In the G_1 and G_0 ODEs, the λ_1 terms reflect that Markov switching from state 1 to state 0 increases G_0 while decreasing G_1 . The $(p_t - G_{1,t})\mu$ term represents the arrival of good evidence, which occurs at rate μ when the state is good and the sender does not already possess good evidence. The $-\kappa_t G_{i,t}$ terms reflect outflows due to disclosure, and the adjustment factor $(1 - G_{1,t} - G_{0,t})$ accounts for the conditioning on no disclosure yet. The p -ODE terms have an analogous interpretation.¹⁰

In the stockpiling phase, $\kappa \equiv 0$, and the system has a closed form solution. The local incentive compatibility condition for delay is

$$\dot{q}_t \geq r q_t - (\lambda_1 + r) p_t. \quad (8)$$

Intuitively, a marginal delay is beneficial when the current flow payoff p_t is large, the posterior after disclosing good evidence q_t is low, or when $\dot{q}_t$ is large (or not too negative). Provided $p_0 > \bar{p}$, the local delay IC condition holds until some time $\underline{T}$, when the sender becomes locally indifferent. This marks the end of the stockpiling phase.

Throughout the purging phase, the IC condition must hold with equality. The system in (G_1, G_0, p, q, κ) is a mixed differential-algebraic system rather than a standard initial value problem (IVP) since κ is not given exogenously, and instead we have an additional equation linking q to G_0 and G_1 . To solve the system, we eliminate G_0 and G_1 using $S := G_0 + G_1$ and $G_1 = qS$, and then we reduce the system to an IVP in two variables (p, q) by solving for S and κ as functions of (p, q) . The resulting equations are (35) and (36) in the appendix,

¹⁰When the bad state is not absorbing, we also must introduce analogous variables $B_{i,t}$ corresponding to bad evidence. The variable B_1 enters the law of motion for p when $\lambda_0 > 0$ since it is possible that the state is good but was bad in the past and the sender already obtained bad evidence, preventing the arrival of good evidence. When $\lambda_0 = 0$, B_1 is identically 0.

and the dynamics of this system are illustrated in Figure 4. The lightly shaded, outlined region, R , is where S and κ are well defined and κ is nonnegative (and therefore a valid arrival rate). In the proof, we show that the disclosure jump $\delta := q_t - p_t$ evolves as

$$\dot{\delta}_t = \delta_t(r + S_t\kappa_t), \quad (9)$$

which implies that sustained indifference requires exponential growth of $q_t - p_t$. Hence, (p, q) must exit this shaded region at some finite time $\bar{T}$, and we show that it can only do so through $q_{\bar{T}} = 1$, marking the end of the purging phase. Hence, as $t \rightarrow \bar{T}$, the purging rate κ_t explodes and the stockpile empties, $S_t \rightarrow 0$.

After the purging phase, the stockpile is empty and the receiver conjectures immediate disclosure at all times. Hence, $q_t = 1$ for all $t \geq \bar{T}$. Since the receiver's belief p_t is below the threshold $\underline{p}$, immediate disclosure is optimal.

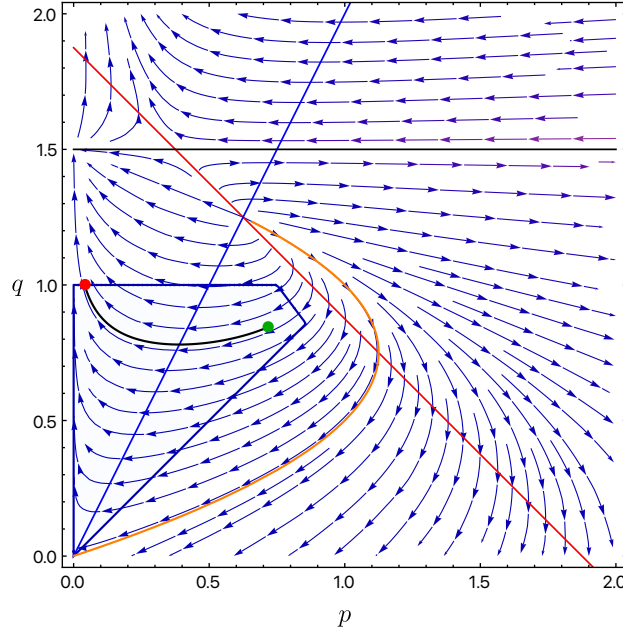


Figure 4: Phase diagram for reduced purging-phase system in (p, q) . The purging phase starts at the green circle and the solution first exits R through $q_{\bar{T}} = 1$ at the red circle. The red and blue lines are isoclines for p and q , respectively, where their derivatives change sign. The ODEs are well defined below the black horizontal line. The orange curve is the set of initial conditions for which the solution converges to the absorbing steady state $(0, 0)$. Parameter values: $\lambda_1 = r = 0.5$, $\mu = 2$, $p_0 = 0.95$.

Robustness. Our restriction to $\lambda_0 = 0$ in the no-timestamps environment is useful as it isolates the analysis of good evidence disclosure by shutting down disclosure of bad evidence.

That said, our results can be extended to $\lambda_0 > 0$. The following result gives a sufficient condition for there to continue to be an equilibrium with no disclosure of bad evidence.

Proposition 4. *For all sufficiently small $\lambda_0 > 0$, there exists an equilibrium with the same three-phase structure for good evidence and no disclosure of bad evidence, supported by beliefs that bad evidence disclosed off-path is fresh.*

4.3 Bad evidence: delay and disclosure bursts

Throughout this section, we assume $\lambda_0 > 0$; if $\lambda_0 = 0$, then disclosing bad evidence cannot be a part of any equilibrium. We focus on equilibria with two properties: (i) age-independent disclosure of bad evidence, and (ii) immediate disclosure of good evidence. Property (i) is natural since the hidden timestamp of bad evidence is not directly payoff relevant to the sender; note that good evidence disclosure in the three-phase equilibrium of Theorem 2 was also age-independent. Property (ii) serves two purposes: first, to isolate the effect of timestamps on bad evidence disclosure, as good evidence is also disclosed immediately in the timestamps environment, and second, to allow us to hold fixed the treatment of good evidence within the no-timestamps environment while varying parameter values to better understand how bad evidence incentives change. To accommodate property (ii), we will study large discount rates, under which, as we will verify, immediate disclosure of good evidence is indeed optimal whenever it is conjectured.¹¹

Since we are now assuming that the bad state is not absorbing, the sender can potentially benefit from disclosing bad evidence because it eliminates the penalty for not disclosing good evidence in the future. However, in contrast to the timestamps environment, the sender cannot prove that old bad evidence is indeed old and wait until disclosure is myopically beneficial; after the disclosure of bad evidence, the receiver updates based on the distribution of arrival dates. With age-independent disclosure, disclosing bad evidence unambiguously and immediately decreases the receiver’s belief about the state. The reason is that bad evidence disclosure proves that the sender indeed had bad evidence, and the receiver’s distribution of arrival dates simply scales up proportionally to account for the mass that was previously assigned to the probability that no evidence has arrived yet.

Thus, the sender faces a trade-off between long-run benefit and short-run cost, and the sender’s optimal disclosure timing depends critically on his discount rate. This is in stark contrast to the timestamps case, where the equilibrium is independent of the discount rate.

The following result says that within our focal class, by taking r sufficiently large, bad

¹¹Immediate disclosure of good evidence would be trivially and uniquely optimal for all discount rates in case $\lambda_1 = 0$, which we have ruled out by assumption.

evidence can only be disclosed after an arbitrarily long delay period. It also establishes that, under a mild restriction on off-path beliefs (and no requirement of age-independent bad evidence disclosure), the only fixed good evidence strategy that would survive for arbitrarily large r is precisely immediate disclosure of good evidence, justifying our focus on equilibria with this strategy. We say that off-path beliefs satisfy the *support restriction* if at each time t , and for each type of evidence, they place zero probability on arrival dates for which disclosure before time t would be expected with probability 1.

Proposition 5. *For any fixed good evidence strategy with positive probability of delay at some time, if r is sufficiently large, it cannot be part of an equilibrium satisfying the support restriction. Moreover, for any $T > 0$, if r is sufficiently large, then in all age-independent equilibria with immediate disclosure of good evidence, there is zero probability of bad evidence disclosure in $[0, T]$.*

Despite the fact that bad evidence is being withheld on $[0, T]$ (for r sufficiently large), effectively pooling bad evidence types with the no-evidence type, the Minimum Principle (Proposition 2) guarantees that at all times, the nondisclosure belief is weakly higher than in the timestamps environment. The reason is that on $[0, T]$, the bad evidence that is disclosed in the timestamps environment is precisely evidence which raises the receiver’s belief (at all future times).

Proposition 5 on its own does not specify how or whether bad evidence is disclosed after time T . It turns out that different off-path beliefs can support different equilibria. Using belief threats that specify any off-path disclosure of bad evidence must be fresh, there exists an equilibrium with no disclosure of bad evidence when r is sufficiently large. However, the following lemma shows that under *neutral* off-path beliefs in the sense of Definition 1, there is no equilibrium in which bad evidence is concealed forever; there is some time T at which the sender would strictly prefer to disclose bad evidence.¹²

Lemma 2. *Under neutral beliefs after off-path disclosure of bad evidence, there is no equilibrium in which good evidence is immediately disclosed and bad evidence is never disclosed. Under threat beliefs, such an equilibrium exists if r is sufficiently large.*

It follows that there cannot be an equilibrium with immediate disclosure of all bad evidence from that point forward; Lemma 3 establishes this formally. In such an equilibrium, all bad evidence disclosed after the first instant would be interpreted as fresh, so after disclosure, the belief would drop to 0 and evolve according to the “no news is no news” Markov switching dynamics. By not disclosing, the belief stays positive and still evolves according to

¹²The neutral belief is the probability the state is good conditional on the sender possessing bad evidence.

those same dynamics: the receiver expects evidence to be disclosed at rate μ in either state, so again no news is no news. Hence, disclosure would result not only in a lower expected payoff but a lower flow payoff at all future times.

Lemma 3. *There is no equilibrium in which good evidence is always disclosed immediately and bad evidence is disclosed immediately at all $t \geq t_0$ for some $t_0 \geq 0$.*

Lemma 3 is superficially similar to the failure of “bang-bang” good evidence disclosure in Lemma 1, but the underlying reason is fundamentally different. There, a complete atom of disclosure gives the sender an incentive to marginally delay, and this dilemma is resolved through continuous mixing prior to the transparency phase. Here, the problem is that an empty stockpile after t_0 reduces incentives for types just after t_0 to disclose, reinforcing the incentive for earlier types to disclose at t_0 itself. A smooth emptying of that stockpile by t_0 through mixing would not resolve this problem.

In sum, the preceding observations suggest equilibria for large r in which good evidence is immediately disclosed, and bad evidence is initially held back, but later released in a pure but *intermittent* fashion. Our main result for bad evidence, Theorem 3, shows that there exists an equilibrium with exactly these properties: bad evidence is disclosed along an infinite sequence of isolated *bursts*. We utilize a stationary prior $p_0 = p^*$, which admits a stationary equilibrium, to simplify the characterization and facilitate comparison to the timestamps environment.

Theorem 3 (Bad evidence in isolated bursts). *Assume $p_0 = p^*$. Then if r is sufficiently large, there exists an equilibrium in which good evidence is disclosed immediately and bad evidence is disclosed at the next time in the set $\{0, \Delta, 2\Delta, \dots\}$, where $\Delta > 0$. Neutral off-path beliefs are sufficient to support this equilibrium.*

The existence of such an equilibrium rests on two key incentive constraints. The first constraint is that in between bursts, if the sender has bad evidence, he must be willing to wait until the next burst; this constraint places an upper bound on Δ .¹³ The second constraint is that the sender must be willing to disclose bad evidence at any given burst rather than wait until the next one; this constraint places a lower bound on Δ . These upper and lower bounds both tend to infinity as $r \rightarrow \infty$, and the proof shows that they have the correct ranking when r is large.

The equilibrium in Theorem 3 exhibits stationary behavior illustrated in Figure 5. The unconditional belief ϕ_t remains constant at p^* by definition. In between bursts, only good

¹³Of course, satisfying this constraint for neutral off-path beliefs means it would also be satisfied for threat beliefs that disclosed bad evidence is fresh.

evidence is disclosed, so no news is bad news, causing the no-disclosure belief to decay, but this decay can be offset eventually by upward mean-reversion due to the hidden Markov switching, resulting in the hook shape. Since each burst results in the disclosure of any possessed evidence, the belief resets to p^* . Meanwhile, the neutral off-path belief resets to 0 after each burst due to the empty stockpile, and it rises as the stock of bad evidence ages.

As a consequence of Lemma 4, which we discuss later, the sender's expected payoff is the same in both the timestamps equilibrium and the burst equilibrium we have constructed. By Proposition 2, the sender's flow payoff prior to any disclosure is higher in the burst equilibrium. Furthermore, his payoff after disclosure of good evidence is identical in both equilibria. It follows that, for each sample path in which good evidence eventually arrives, the sender prefers the burst equilibrium. Indifference then implies that averaging over the paths in which bad evidence eventually arrives, the sender prefers the timestamps equilibrium.

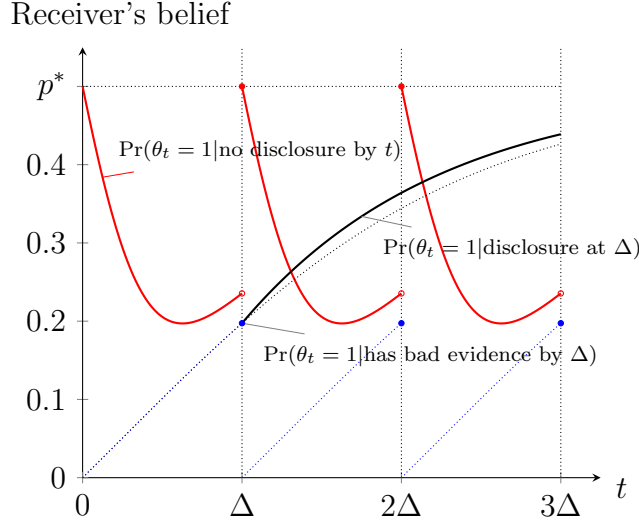


Figure 5: The receiver's beliefs in a stationary burst equilibrium. The red curve shows the belief following no disclosure, the dotted black curve shows the off-path belief following disclosure between bursts, and the solid black curve shows the belief following disclosure at the burst time Δ . The parameters are $\lambda_0 = 1, \lambda_1 = 1, \mu = 10, p_0 = p^* = 1/2, r = 5$, and $\Delta = 0.4$.

5 Discussion

5.1 Receiver behavior and welfare

Although we do not establish uniqueness within the general class of equilibria, the sender's expected payoff is the same in all equilibria, given the parameter values. More strongly, the

following result establishes that as long as the receiver’s conjecture is correct (independent of sender optimality), the sender’s expected payoff is the same, and thus the sender also cannot benefit from commitment.

Lemma 4. *Fix parameter values. Fix any strategy of the sender, and let $(p_t)_{t \geq 0}$ denote the receiver’s posterior belief process when that strategy is followed and the receiver’s conjecture is correct. Then for all $t \geq 0$, $\mathbb{E}[p_t] = \phi_t = p^* + (p_0 - p^*)e^{-(\lambda_0 + \lambda_1)t}$, which is independent of the sender’s strategy.*

The proof is simply the following: by definition, p_t is the receiver’s time- t expectation of the state θ_t , and by the law of iterated expectations, $\mathbb{E}[p_t] = \mathbb{E}[\theta_t] = \phi_t$, the receiver’s belief based only on knowledge of the Markov switching dynamics.

The sender’s flow payoff p_t in our model is reduced form and can be microfounded as follows. Consider a market of identical receivers $i \sim U[0, 1]$ who, at each instant t , choose actions $a_{i,t}$ and receive payoffs $-\int_0^\infty e^{-rt}(a_{i,t} - \theta_t)^2 dt$. The sender’s flow payoff is $\int_0^1 a_{i,t} di$. Then $a_{i,t} = p_t$ for all (i, t) , and the sender’s flow payoff is exactly p_t as in our model. We then use “receiver” to refer to a representative receiver.

This microfoundation allows us to analyze the effects of timestamps on receiver welfare. Suppose $\lambda_0 = 0$ and consider the equilibrium of Theorem 2. Compared to the unique equilibrium under timestamps, there is delay of good evidence disclosure whenever $p > \underline{p}$, which strictly reduces receiver welfare. Moreover, when $\lambda_0 > 0$ is sufficiently small for Proposition 4, the comparison is exacerbated: bad evidence is not disclosed, while it is always eventually disclosed under timestamps. In addition, for stationary prior and large r , the receiver is also worse off in the burst equilibrium (and in any age-independent equilibrium involving immediate disclosure of good evidence) compared to the timestamps equilibrium, since delay of bad evidence disclosure in the former can be made arbitrarily long by Proposition 5, while it occurs after delay α^* (constant in both time and r) in the latter. These comparisons highlight a negative indirect effect of removing timestamps, in addition to the direct effect.

5.2 Modeling assumptions

Our model assumes the state evolves over time. Holding fixed the rest of the model, this assumption is in fact necessary for timestamps to play a nontrivial role; with a constant state, the sender would immediately disclose good evidence and never disclose bad evidence, for all values of the other parameters.

We assume that evidence only arrives once. This is appropriate in settings where hard evidence is not continuously available, either because producing multiple pieces of evidence is infeasible or too costly. Nonetheless, the proof of Proposition 3 goes through for multiple

pieces of evidence; we establish this for an extension with an unlimited number of pieces of evidence in the Online Appendix. We also conjecture that good evidence would continue to be disclosed immediately under timestamps. We conjecture that bad evidence would still be disclosed if there are only finitely many pieces of evidence, or if the evidence arrival rate decreases sufficiently with each piece. Fully solving such a model would be more technically challenging as the receiver would have to maintain beliefs about a higher-dimensional evidence status of the sender.

We assume that the sender has no private information about the state other than the evidence she obtains. Since the sender’s preferences are state-independent, equilibria in our model would also be equilibria in a model where the sender observes the state at all times. The latter model would allow for stronger off-path belief threats; for example, the receiver could assume that if good evidence is unexpectedly disclosed, the sender must know that the state is *currently* bad.

We assume that the only information available to the receiver (other than time itself and knowledge of the hidden Markov switching) is disclosure by the sender. If the model included exogenous public news, the evidence’s timestamp would help the sender forecast continuation payoffs, changing the disclosure incentives.

A Proofs

A.1 Proof of Theorem 1

The proof proceeds in the following main steps. First, we show that in any equilibrium, good evidence is disclosed immediately (Lemma 5). Second, we show that bad evidence with any timestamp τ is disclosed after some (possibly infinite) time $D(\tau)$ increasing in τ (Corollary 2). Third, we characterize $D(\tau)$ uniquely.

For what follows, recall that $q_t^{G,\tau}$ and $q_t^{B,\tau}$ denote the belief after disclosure at time t of good and bad evidence, respectively, with timestamp τ . Their closed-form expressions are given by (1) and are continuous in t and τ . Moreover, for a fixed t , $q_t^{G,\tau}$ is strictly increasing in τ , while $q_t^{B,\tau}$ is strictly decreasing in τ . Fix an arbitrary conjecture about the sender’s strategy, and let p be the belief path conditional on no disclosure.

A.1.1 Immediate disclosure of good evidence with timestamps

Lemma 5. *In any equilibrium, if the sender possesses good evidence at any time, he discloses it immediately.*

In the rest of this section, we prove Lemma 5.

Lemma 6. *If $p_t < q_t^{G,0}$ for all times, then immediate disclosure of good evidence is optimal regardless of its timestamp.*

Proof. Suppose the sender has good evidence with timestamp τ and hasn't disclosed it before time t . If the sender discloses at time $t' \geq t$, her flow payoff is p_s for $s \in [t, t')$ and $q_s^{G,\tau} \geq q_s^{G,0} > p_s$ for all $s \in [t', \infty)$. Immediate disclosure maximizes the flow payoff at all times. $\square$

Lemma 7. *If the sender possesses bad evidence at any time t and $p_t > q_t^{B,0}$, then there exists $\varepsilon > 0$ such that the sender does not disclose it before time $t + \varepsilon$.*

Proof. Suppose the sender has bad evidence with timestamp τ and hasn't disclosed it before time t . Since p and q^B are right-continuous, there exists $\varepsilon > 0$ such that $p_s > q_s^{B,0} \geq q_s^{B,\tau}$ for all $s \in [t, t + \varepsilon)$. Hence, by disclosing at time $t + \varepsilon$ instead of t , the flow payoff is strictly higher on $[t, t + \varepsilon)$ and is unchanged on $[t + \varepsilon, \infty)$. $\square$

Define $t^G = \inf\{t \geq 0 : p_t \geq q_t^{G,0}\}$ and $t^B = \inf\{t \geq 0 : p_t \leq q_t^{B,0}\}$. By right-continuity of p_t , $q_t^{G,0}$, and $q_t^{B,0}$ in t , if $p_0 \in (0, 1)$, then $t^G, t^B > 0$. Right-continuity also implies $p_{t^G} \geq q_{t^G}^{G,0}$ if t^G is finite, and likewise $p_{t^B} \leq q_{t^B}^{B,0}$ if t^B is finite.

We now show that $t^G = \infty$. Clearly, since $q_t^{G,0} \neq q_t^{B,0}$ for all t , we cannot have $t^G = t^B < \infty$. Hence, if $t^G < \infty$, we have either $t^G < t^B$ or $t^B < t^G$.

The following lemma establishes that if t^G is finite, p must jump above $q^{G,0}$ from below $q^{B,0}$ at time t^G .

Lemma 8. *If $t^G < \infty$, then $t^B < t^G$ and $\sup\{t < t^G : p_t \leq q_t^{B,0}\} = t^G$.*

Proof. Toward a contradiction, suppose there exists $\varepsilon > 0$ such that for all $t \in [t^G - \varepsilon, t^G)$, $p_t > q_t^{B,0}$. By right-continuity, we can choose this ε such that $p_t > q_t^{B,0}$ for all $t \in [t^G - \varepsilon, t^G + \varepsilon)$. To simplify notation, set $t_0 := t^G - \varepsilon$ and $t_1 := t^G + \varepsilon$. Then by Lemma 7, there is no disclosure of bad evidence on $[t_0, t_1)$. Hence, on $[t_0, t_1)$, p is bounded above by the solution to $\dot{y}_t = \lambda_0(1 - y_t) - \lambda_1 y_t$ with initial condition $y_{t_0} = p_{t_0}$. That is, $y_t = \mathbb{P}_{t_0}(\theta_t = 1)$ is the unconditional belief based on Markov switching, starting from p_{t_0} . To prove this, let $\mathbb{E}_t$ denote the receiver's expectation operator at time t , and $\mathbb{P}_t$ the probability measure at time t . By the law of iterated expectations, for all $t \in [t_0, t_1)$,

$$y_t = \mathbb{P}_{t_0}(\theta_t = 1) = \mathbb{P}_{t_0}(\text{disclosure by } t) \mathbb{P}_{t_0}(\theta_t = 1 | \text{disclosure by } t) + \mathbb{P}_{t_0}(\text{no disclosure by } t) p_t.$$

Because $p_t > q_t^{B,0}$, no bad evidence is disclosed in $[t_0, t_1)$, so any disclosure is of good evidence. The lowest post-disclosure belief from good evidence is $q_t^{G,0}$; good evidence with any later timestamp yields a strictly higher belief. So $\mathbb{P}_{t_0}(\theta_t = 1 | \text{disclosure by } t) \geq q_t^{G,0}$.

Moreover, $y_t < q_t^{G,0}$. This is because y and q^G satisfy the same linear ODE on $[t_0, t_1)$ and differ only in their initial value; we have $q_{t_0}^{G,0} > y_{t_0}$, so by the comparison theorem, $y_t < q_t^{G,0}$ for all $t \geq t_0$.

Putting these facts together, we have

$$\begin{aligned} y_t &= \mathbb{P}_{t_0}(\text{disclosure by } t) \mathbb{P}_{t_0}(\theta_t = 1 | \text{disclosure by } t) + \mathbb{P}_{t_0}(\text{no disclosure by } t) p_t \\ &\geq \mathbb{P}_{t_0}(\text{disclosure by } t) q_t^{G,0} + \mathbb{P}_{t_0}(\text{no disclosure by } t) p_t \\ &> \mathbb{P}_{t_0}(\text{disclosure by } t) y_t + \mathbb{P}_{t_0}(\text{no disclosure by } t) p_t. \end{aligned}$$

Since $\mathbb{P}_{t_0}(\text{no disclosure by } t) > 0$, the last inequality implies that $p_t < y_t < q_t^{G,0}$ for all $t \in [t_0, t_1)$, contradicting the definition of t^G . $\square$

Lemma 9. *It must be that $t^G = \infty$.*

Proof. By Lemma 8, if $t^G < \infty$, then at time t , p must jump from weakly below $q^{B,0}$ to weakly above $q^{G,0}$. Such a jump requires that good or bad evidence is conjectured to be disclosed with an atom of probability at time t^G . Now if only an atom of good evidence disclosure is conjectured, p would exhibit a downward discontinuity, since any good evidence disclosure would raise the belief given that $p_{t^G-} \leq q_{t^G}^{B,0} < q_{t^G}^{G,0}$. Thus, at time t^G , the receiver must conjecture that bad evidence is disclosed with an atom of probability. But then disclosing bad evidence at time t^G would not be optimal, as there exists $\varepsilon > 0$ such that $p_{t^G} \geq q_{t^G}^{G,0} > q_{t^G}^{B,0} \geq q_{t^G}^{B,\tau}$ for all $t \in [t^G, t^G + \varepsilon)$ and all $\tau \in [0, t^G]$. In other words, the sender would strictly prefer to wait and disclose at time $t^G + \varepsilon$. $\square$

Since $t^G = \infty$, we have $p_t < q_t^{G,0}$ for all $t \geq 0$, and thus it is optimal to disclose good evidence immediately.

A.1.2 Delayed disclosure of bad evidence with timestamps

Lemma 10. *In equilibrium, p is continuous.*

Proof. If p has a discontinuity at some time t_0 , then it must be that at time t_0 , evidence is disclosed with an atom of probability. Since good evidence is disclosed immediately by Lemma 5 and arrives according to an atomless distribution, it must be that bad evidence is disclosed with an atom of probability at time t_0 . Note that for any type τ_0 with bad evidence that is willing to disclose at t_0 , it must be that $q_{t_0}^{B,\tau_0} \geq p_{t_0}$, otherwise by right-continuity of p and continuity of q^{B,τ_0} we would have $p_t > q_t^{B,\tau_0}$ for all $t \in [t_0, t_0 + \varepsilon)$ for some $\varepsilon > 0$, and the sender would strictly prefer to disclose at $t_0 + \varepsilon$.

We now argue that if there is a discontinuity at t_0 , it must be that $p_{t_0} < p_{t_0-}$, where the left limit exists because p is càdlàg. For any $t < t_0$, let N and G denote the events that there is no disclosure or there is good evidence disclosure in $[t, t_0]$ (suppressing dependence on t and t_0). Let B^{t_0} denote the event that there is bad evidence disclosure at time t_0 , and $B^{[t, t_0]}$ the event that there is bad evidence disclosure in $[t, t_0]$. Let $\mathbb{E}_t$ and $\mathbb{P}_t$ denote expectations and probabilities conditional on no disclosure prior to t . By the law of iterated expectations,

$$\mathbb{E}_t[\theta_{t_0}] = \mathbb{P}_t(N)p_{t_0} + \mathbb{P}_t(G)\mathbb{E}_t[\theta_{t_0}|G] + \mathbb{P}_t(B^{[t, t_0]})\mathbb{E}_t[\theta_{t_0}|B^{[t, t_0]}] + \mathbb{P}_t(B^{t_0})\mathbb{E}_t[\theta_{t_0}|B^{t_0}], \quad (10)$$

where $\mathbb{E}_t[\theta_{t_0}|B^{t_0}] \geq p_{t_0}$ by the argument above. As $t \uparrow t_0$, $\mathbb{P}_t(G) \rightarrow 0$ and $\mathbb{P}_t(B^{[t, t_0]}) \rightarrow 0$, so $\mathbb{P}_t(N) + \mathbb{P}_t(B^{t_0}) \rightarrow 1$. To simplify notation, define $\alpha := \lim_{t \uparrow t_0} \mathbb{P}_t(N)$, $\beta := \lim_{t \uparrow t_0} \mathbb{P}_t(B^{t_0}) = 1 - \alpha$, and $\gamma := \lim_{t \uparrow t_0} \mathbb{E}_t[\theta_{t_0}|B^{t_0}] \geq p_{t_0}$. Then $\alpha > 0$ because the probability of no disclosure in $[t, t_0]$ is at least the probability that evidence arrives after time t_0 , which is strictly positive; moreover, $\beta > 0$ because there is an atom of bad evidence disclosure at time t_0 .

Thus, the limit of the right-hand side of (10) as $t \uparrow t_0$ is $\alpha p_{t_0} + \beta \gamma \geq \alpha p_{t_0} + \beta p_{t_0} = p_{t_0}$. The limit of the left-hand side of (10) is p_{t_0-} . It follows that $p_{t_0-} \geq p_{t_0}$. A discontinuity at t_0 must therefore imply that $p_{t_0-} > p_{t_0}$. This further implies $\gamma = \frac{p_{t_0-} - \alpha p_{t_0}}{\beta} = \frac{p_{t_0-} - \alpha p_{t_0}}{1 - \alpha} > p_{t_0-}$.

Since $\lim_{t \uparrow t_0} \mathbb{E}_t[\theta_{t_0}|B^{t_0}] = \gamma > p_{t_0-}$, we have $\mathbb{E}_t[\theta_{t_0}|B^{t_0}] > p_{t_0-}$ for t sufficiently close to t_0 . As $\mathbb{E}_t[\theta_{t_0}|B^{t_0}]$ is a convex combination over τ of $q_{t_0}^{B, \tau}$ values, there exists some timestamp $\tau < t_0$ for bad evidence for which the sender is willing to wait until t_0 to disclose and $q_{t_0}^{B, \tau} > p_{t_0-}$. By definition of p_{t_0-} , for any $\delta > 0$, there exists $\varepsilon > 0$ such that if $t > t_0 - \varepsilon$, $|p_t - p_{t_0-}| < \delta$. Choosing δ sufficiently small and ε accordingly, we have $p_t < q_t^{B, \tau}$ for all $t \in (t_0 - \varepsilon, t_0)$. But then the sender would strictly prefer to disclose at any such t rather than wait until time t_0 , contradicting optimality. $\square$

Lemma 11 (Single-crossing). *If $p_{t_0} \leq q_{t_0}^{B, \tau_0}$ for some $0 \leq \tau_0 \leq t_0$, then $p_t < q_t^{B, \tau_0}$ for all $t > t_0$.*

Proof. Assume $p_{t_0} \leq q_{t_0}^{B, \tau_0}$ for some $0 \leq \tau_0 \leq t_0$. Define $t^* = \inf\{t > t_0 : p_t \geq q_t^{B, \tau_0}\}$. We aim to show that $t^* = \infty$. First, we rule out $t^* \in (t_0, \infty)$, and then we rule out $t^* = t_0$. Toward a contradiction, suppose $t^* \in (t_0, \infty)$.

For each $\tau \leq t^*$, define $F(\tau) = \max_{t \in [\max\{t_0, \tau\}, t^*]} (q_t^{B, \tau} - p_t)$. Since q^B is continuous in both arguments and p is continuous by Lemma 10, F is continuous by the maximum theorem. Moreover, F is strictly decreasing since q^B is strictly decreasing in τ and the set $[\max\{t_0, \tau\}, t^*]$ is weakly decreasing in τ in the set inclusion sense; it satisfies $F(\tau_0) > 0$ since $t^* > t_0$ by assumption, and by the definition of t^* we cannot have $p_t \geq q_t^{B, \tau_0}$ for any $t \in (t_0, t^*)$; and it satisfies $F(t^*) < 0$ since $q_{t^*}^{B, t^*} = 0 < p_{t^*}$. Thus, there exists $\bar{\tau}$ such that

$F(\tau) \geq 0$ if and only if $\tau \leq \bar{\tau}$. Since $F(\bar{\tau}) = 0$, there exists some $t_1 \in [\max\{t_0, \bar{\tau}\}, t^*)$ where $q_{t_1}^{B, \bar{\tau}} = p_{t_1}$, with $q_t^{B, \bar{\tau}} \leq p_t$ for all $t \in [\max\{t_0, \bar{\tau}\}, t^*]$.

Define $\phi_t(p_{t_1}, t_1)$ as the solution to $\dot{y}_t = \lambda_0(1 - y_t) - \lambda_1 y_t$ subject to the initial condition $y_{t_1} = p_{t_1}$. Note that q_t^{B, τ_0} solves the same ODE. Since $p_{t_1} < q_{t_1}^{B, \tau_0}$, it follows that $\phi_{t^*}(p_{t_1}, t_1) < q_{t^*}^{B, \tau_0} = p_{t^*}$. Now let N , G , and B denote the events that no evidence, good evidence, and bad evidence, respectively, is disclosed in $[t_1, t^*]$. By the law of iterated expectations,

$$\mathbb{E}_{t_1}[\theta_{t^*}] = \phi_{t^*}(p_{t_1}, t_1) = \mathbb{P}_{t_1}(N)p_{t^*} + \mathbb{P}_{t_1}(G)\mathbb{E}_{t_1}[\theta_{t^*}|G] + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{t^*}|B].$$

Now $\mathbb{E}_{t_1}[\theta_{t^*}|G] > q_{t^*}^{B, \tau_0} = p_{t^*}$. Hence,

$$\phi_{t^*}(p_{t_1}, t_1) \geq (1 - \mathbb{P}_{t_1}(B))p_{t^*} + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{t^*}|B].$$

Recall that $p_{t^*} > \phi_{t^*}(p_{t_1}, t_1)$, and $\mathbb{P}_{t_1}(B) < 1$, so we must have

$$\begin{aligned} \phi_{t^*}(p_{t_1}, t_1) &> (1 - \mathbb{P}_{t_1}(B))\phi_{t^*}(p_{t_1}, t_1) + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{t^*}|B] \\ \implies \phi_{t^*}(p_{t_1}, t_1) &> \mathbb{E}_{t_1}[\theta_{t^*}|B], \end{aligned}$$

where $\mathbb{P}_{t_1}(B) > 0$ as the first inequality cannot hold if $\mathbb{P}_{t_1}(B) = 0$. Since $\mathbb{E}_{t_1}[\theta_{t^*}|B] = \mathbb{E}_{t_1}[q_{t^*}^{B, \tau}|B]$, this means there must be some timestamp τ and some time $t \in (t_1, t^*)$ for which bad evidence is disclosed at time t and $q_t^{B, \tau} < \phi_{t^*}(p_{t_1}, t_1)$. Since $\phi_{t^*}(p_{t_1}, t_1) = q_{t^*}^{B, \bar{\tau}}$ on $[t_1, t^*]$ (as $q_{t^*}^{B, \bar{\tau}}$ solves the same ODE with same initial value $q_{t_1}^{B, \bar{\tau}} = p_{t_1} = \phi_{t_1}(p_{t_1}, t_1)$), we have $q_{t^*}^{B, \tau} < q_{t^*}^{B, \bar{\tau}}$, and because $q_s^{B, \cdot}$ is strictly decreasing in the timestamp on $[0, s]$ for each fixed s , this implies $\tau > \bar{\tau}$. But then we have $F(\tau) < 0$, so $p_t > q_t^{B, \tau}$ for all $t \in (\max\{t_0, \tau\}, t^*)$, and this implies that the sender would strictly prefer to delay the disclosure until time t^* , a contradiction.

Next, we rule out the case $t^* = t_0$ by adapting arguments from the previous case. Toward a contradiction, suppose that $t^* = t_0$. Then for all $\varepsilon > 0$ there exists $\hat{t} \in (t_0, t_0 + \varepsilon)$ such that $p_{\hat{t}} \geq q_{\hat{t}}^{B, \tau_0}$. By arguments similar to those used in the previous case, there exists some timestamp $\bar{\tau} \geq \tau_0$ and time $t_1 \in [\max\{\bar{\tau}, t_0\}, \hat{t})$ such that $p_{t_1} = q_{t_1}^{B, \bar{\tau}}$ and $p_t \geq q_t^{B, \bar{\tau}}$ for all $t \in [\max\{\bar{\tau}, t_0\}, \hat{t}]$. Since $\phi_{t^*}(p_{t_1}, t_1)$ and $q_{t^*}^{B, \bar{\tau}}$ solve the same ODE and coincide at t_1 , we have $\phi_{t^*}(p_{t_1}, t_1) = q_{t^*}^{B, \bar{\tau}}$ for all $s \geq t_1$, and in particular for $s = \hat{t}$.

Let N , G , and B be the events of no disclosure, good evidence disclosure, and bad evidence disclosure on $[t_1, \hat{t}]$. By similar logic to that used in the proof for the $t^* \in (t_0, \infty)$ case, we have $\phi_{\hat{t}}(p_{t_1}, t_1) = q_{\hat{t}}^{B, \bar{\tau}} \leq q_{\hat{t}}^{B, \tau_0} \leq p_{\hat{t}}$ and also $\mathbb{E}_{t_1}[\theta_{\hat{t}}|G] > q_{\hat{t}}^{B, \tau_0} \geq q_{\hat{t}}^{B, \bar{\tau}}$. Thus, by the

law of iterated expectations,

$$\begin{aligned}\mathbb{E}_{t_1}[\theta_{\hat{t}}] &= q_{\hat{t}}^{B,\bar{\tau}} = \mathbb{P}_{t_1}(N)p_{\hat{t}} + \mathbb{P}_{t_1}(G)\mathbb{E}_{t_1}[\theta_{\hat{t}}|G] + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{\hat{t}}|B] \\ &> \mathbb{P}_{t_1}(N)q_{\hat{t}}^{B,\bar{\tau}} + \mathbb{P}_{t_1}(G)q_{\hat{t}}^{B,\bar{\tau}} + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{\hat{t}}|B].\end{aligned}$$

As in the previous case, the strict inequality implies $\mathbb{P}_{t_1}(B) > 0$. Thus,

$$q_{\hat{t}}^{B,\bar{\tau}} > \mathbb{E}_{t_1}[\theta_{\hat{t}}|B].$$

Since the expectation is a convex combination of $q_{\hat{t}}^{B,\tau}$ values over $\tau \leq \hat{t}$, there must be disclosure at some time $t \in [t_1, \hat{t}]$ of bad evidence with some timestamp $\tau > \bar{\tau}$. But at t , we have $p_t \geq q_t^{B,\bar{\tau}} > q_t^{B,\tau}$. By continuity in t , this inequality holds for all times in an interval $[t, t + \varepsilon)$ for some $\varepsilon > 0$, and the sender would strictly prefer to delay disclosure until $t + \varepsilon$, contradicting optimality. This rules out $t^* = t_0$. $\square$

Corollary 2 (Deterministic delays). *In equilibrium, bad evidence with timestamp τ is disclosed at some time $D(\tau) \in (\tau, \infty]$. Moreover, there exists $\bar{\tau} \in (0, \infty]$ such that for $\tau \in [0, \bar{\tau})$, $D(\tau)$ is strictly increasing and finite, and for $\tau \geq \bar{\tau}$, $D(\tau) = \infty$.*

Proof. Suppose that at some time t_0 , the sender has bad evidence with timestamp $\tau_0 \leq t_0$. Let $D(\tau_0, t_0) := \inf\{t > t_0 : p_t \leq q_t^{B,\tau_0}\}$. (Note that $D(\tau_0, t_0) > \tau_0$, since $q_{\tau_0}^{B,\tau_0} = 0 < p_{\tau_0}$.) We claim that disclosing at time $D(\tau_0, t_0)$ is optimal. If $D(\tau_0, t_0) = \infty$, then we have $p_t > q_t^{B,\tau_0}$ for all $t \geq t_0$, and clearly it is optimal for the sender to never disclose the evidence. If $D(\tau_0, t_0) < \infty$, then we have (i) $p_t > q_t^{B,\tau_0}$ for all $t < D(\tau_0, t_0)$, which implies disclosing at time $D(\tau_0, t_0)$ yields a higher payoff than disclosing at any earlier time, and (ii) $p_t < q_t^{B,\tau_0}$ for all $t > D(\tau_0, t_0)$, which implies that disclosing at time $D(\tau_0, t_0)$ yields a higher payoff than disclosing at any later time. On path, it is therefore optimal to disclose at time $D(\tau) := D(\tau, \tau)$.

Define $\bar{\tau} := \inf\{\tau \geq 0 : D(\tau) = \infty\}$. Since $q_t^{B,\tau}$ is strictly increasing in t and strictly decreasing in τ , we have that $D(\tau)$ is increasing on $[0, \infty)$ and strictly increasing on $[0, \bar{\tau})$. $\square$

We now turn to characterizing $\tau \mapsto D(\tau)$. Note that, in equilibrium, when $p_t = q_t^{B,s}$ for some s , p_t admits a direct characterization via Bayes' rule:

$$p_t = p_{t,s}^\dagger := \frac{e^{-\mu t}\phi_t + \int_s^t \mu e^{-\mu\tau}(1 - \phi_\tau)q_t^{B,\tau}d\tau}{e^{-\mu t} + \int_s^t \mu e^{-\mu\tau}(1 - \phi_\tau)d\tau}. \quad (11)$$

As mentioned in the main text, $p_{t,s}^\dagger$ is an auxiliary belief process that specifies the probability that $\theta_t = 1$ conditional on no disclosure up to time t , under a conjecture that (i) good

evidence is always disclosed immediately and (ii) bad evidence with timestamp before s is disclosed before t , and no evidence with timestamp after s is disclosed before t . We show that $p_{t^\dagger(s)} = p_{t^\dagger(s),s}^\dagger$ for all $s \geq 0$.¹⁴

Lemma 12. *If $\lambda_0, \lambda_1 > 0$, then for all $s \geq 0$, there exists $t^\dagger(s)$ that solves (4). Moreover, $t^\dagger(s)$ is unique and lies in (s, ∞) .*

Proof. Expanding (4) and multiplying through by the denominator yields

$$\begin{aligned} e^{-\mu t} \phi_t + \int_s^t \mu e^{-\mu \tau} (1 - \phi_\tau) q_t^{B,\tau} d\tau &= q_t^{B,s} \left[e^{-\mu t} + \int_s^t \mu e^{-\mu \tau} (1 - \phi_\tau) d\tau \right] \\ \iff \phi_t - q_t^{B,s} &= \int_s^t \mu e^{-\mu(\tau-t)} (1 - \phi_\tau) (q_t^{B,s} - q_t^{B,\tau}) d\tau. \end{aligned} \quad (12)$$

We establish existence by the intermediate value theorem. At $t = s$, the left-hand side is $\phi_s - 0 > 0$, while the right-hand side vanishes. As $t \rightarrow \infty$, the left-hand side tends to $p^* - p^* = 0$. Meanwhile, for fixed $\varepsilon > 0$, the right-hand side is bounded below by

$$\int_{t-\varepsilon}^t \mu e^{-\mu(\tau-t)} (1 - \phi_\tau) (q_t^{B,s} - q_t^{B,\tau}) d\tau \geq \int_{t-\varepsilon}^t \mu e^{-\mu\varepsilon} (1 - \phi_\tau) (q_t^{B,s} - q_t^B(t-\varepsilon)) d\tau.$$

As $t \rightarrow \infty$, we have $\phi_t \rightarrow p^* < 1$ under the assumption that $\lambda_1 > 0$, and we also have $q_t^{B,s} \rightarrow p^* > 0$ under the assumption that $\lambda_0 > 0$. In addition, $q_t^{B,t-\varepsilon} < p^*(\lambda_0 + \lambda_1)\varepsilon$. Hence, for small ε and large t the integrand is strictly positive, and we conclude that the right-hand side of (12) is positive for large t . Since both the left- and right-hand side of (12) are continuous in t , there exists a solution $t^\dagger(s) \in (s, \infty)$ (and no solution at the ends of this interval).

Uniqueness follows from showing that $p_{t,s}^\dagger$ crosses $q_t^{B,s}$ from above at $t^\dagger(s)$. Given Markov switching, $\dot{q}_t^{B,s} = \lambda_0(1 - q_t^{B,s}) - \lambda_1 q_t^{B,s}$. For fixed s , we can write the evolution of $p_{t,s}^\dagger$ as

$$\dot{p}_{t,s}^\dagger = \lambda_0(1 - p_{t,s}^\dagger) - \lambda_1 p_{t,s}^\dagger - \frac{\mu e^{-\mu t} \phi_t}{e^{-\mu t} + \int_s^t \mu e^{-\mu \tau} (1 - \phi_\tau) d\tau} (1 - p_{t,s}^\dagger). \quad (13)$$

At $t^\dagger(s)$, $p_{t^\dagger(s),s}^\dagger = q_{t^\dagger(s)}^{B,s}$, so

$$\dot{q}_{t^\dagger(s)}^{B,s} - \dot{p}_{t^\dagger(s),s}^\dagger = \frac{\mu e^{-\mu t^\dagger(s)} \phi_{t^\dagger(s)}}{e^{-\mu t^\dagger(s)} + \int_s^{t^\dagger(s)} \mu e^{-\mu \tau} (1 - \phi_\tau) d\tau} (1 - p_{t^\dagger(s),s}^\dagger) > 0.$$

The result follows. □

¹⁴For fixed s , $p_{t,s}^\dagger$ does not coincide at all times with the equilibrium belief process p_t , as these belief processes are based on different conjectures about the disclosure of bad evidence.

A.2 Proof of Proposition 1

By definition, $t^\dagger(s) = s + a^*(s)$. Recall the equilibrium condition $q_{s+a^*(s)}^{B,s} = p_{s+a^*(s),s}^\dagger$. The equilibrium belief following non-disclosure by time t , denoted p_t , is equal to $p_{t,0}^\dagger$ if $t \leq a^*(0)$ and $p_{s+a^*(s),s}^\dagger$ if $t \geq a^*(0)$ where $s + a^*(s) = t$.

We first show $a^*(s)$ is differentiable in s . Define

$$H(s, a) := q_{s+a}^{B,s} - p_{s+a,s}^\dagger.$$

By Lemma 12, $a^*(s)$ is the unique solution to $H(s, a) = 0$ for each s . By the implicit function theorem, $a^*(s)$ is continuously differentiable at each $s \geq 0$.

At $t = s + a^*(s)$, $\partial q_t^{B,s}/\partial s = -\partial q_t^{B,s}/\partial t$ and $\partial p_{t,s}^\dagger/\partial s = 0$, where the second equality follows from $p_{t,s}^\dagger = q_t^{B,s}$. Differentiating the equilibrium condition $q_{s+a^*(s)}^{B,s} = p_{s+a^*(s),s}^\dagger$ implicitly in s and using the above equalities, we obtain

$$\frac{da^*(s)}{ds} = \frac{\partial p_{t,s}^\dagger/\partial t}{\partial q_t^{B,s}/\partial t - \partial p_{t,s}^\dagger/\partial t} \Big|_{t=s+a^*(s)} \quad \text{and} \quad 1 + \frac{da^*(s)}{ds} = \frac{\partial q_t^{B,s}/\partial t}{\partial q_t^{B,s}/\partial t - \partial p_{t,s}^\dagger/\partial t} \Big|_{t=s+a^*(s)}. \quad (14)$$

We have shown that $\partial q_t^{B,s}/\partial t - \partial p_{t,s}^\dagger/\partial t > 0$, so $1 + da^*(s)/ds > 0$. That is, $s + a^*(s)$ is increasing in s for $s \geq 0$ and strictly increasing if $s > 0$. This proves (i).

From (14), $da^*(s)/ds$ has the same sign as $\partial p_{t,s}^\dagger/\partial t|_{t=s+a^*(s)}$. In what follows, we derive necessary and sufficient conditions under which $\partial p_{t,s}^\dagger/\partial t|_{t=s+a^*(s)}$ is (strictly) positive.

Let $t = s + a$ with $a \geq 0$ fixed. Then

$$\frac{dp_{s+a,s}^\dagger}{ds} = \frac{\partial p_{t,s}^\dagger}{\partial t} \Big|_{t=s+a} + \frac{\partial p_{t,s}^\dagger}{\partial s} \Big|_{t=s+a}.$$

As shown before, $\frac{\partial p_{t,s}^\dagger}{\partial s} \Big|_{t=s+a^*(s)} = 0$. So at $a = a^*(s)$,

$$\frac{dp_{s+a,s}^\dagger}{ds} \Big|_{a=a^*(s)} = \frac{\partial p_{t,s}^\dagger}{\partial t} \Big|_{t=s+a^*(s)}.$$

Then, it suffices to derive necessary and sufficient conditions under which $dp_{s+a,s}^\dagger/ds|_{t=s+a^*(s)}$ is (strictly) positive for any $s \geq 0$. By definition,

$$p_{s+a,s}^\dagger = \frac{e^{-\mu(s+a)}\phi_{s+a} + \int_s^{s+a} \mu e^{-\mu\tau}(1 - \phi_\tau)q_{s+a}^{B,\tau}d\tau}{e^{-\mu(s+a)} + \int_s^{s+a} \mu e^{-\mu\tau}(1 - \phi_\tau)d\tau}.$$

Via a change of variable, let $\tau = s + v$ where $v \in [0, a]$. Then $d\tau = dv$ (since s is fixed).

After some simplifying, we get

$$p_{s+a,s}^\dagger = \frac{e^{-\mu a} \phi_{s+a} + \int_0^a \mu e^{-\mu v} (1 - \phi_{s+v}) q_{s+a}^{B,s+v} dv}{e^{-\mu a} + \int_0^a \mu e^{-\mu v} (1 - \phi_{s+v}) dv}. \quad (15)$$

Note that $q_{s+a}^{B,s+v} = p^* (1 - e^{-(\lambda_0 + \lambda_1)(a-v)})$ is independent of s . This means $p_{s+a,s}^\dagger$ changes with s only through its changes in ϕ . Differentiate both sides with respect to s and note that $d\phi_{s+x}/ds = d\phi_t/dt|_{t=s+x} =: \dot{\phi}_{s+x}$. After simplifying,

$$\frac{dp_{s+a,s}^\dagger}{ds} = \frac{e^{-\mu a} \dot{\phi}_{s+a} + \int_0^a \mu e^{-\mu v} \dot{\phi}_{s+v} (p_{s+a,s}^\dagger - q_{s+a}^{B,s+v}) dv}{e^{-\mu a} + \int_0^a \mu e^{-\mu v} (1 - \phi_{s+v}) dv}.$$

Evaluate at $a = a^*(s)$. Because the denominator is positive, $\frac{dp_{s+a,s}^\dagger}{ds}|_{a=a^*(s)}$ has the same sign as the numerator:

$$e^{-\mu a^*(s)} \dot{\phi}_{s+a^*(s)} + \int_0^{a^*(s)} \mu e^{-\mu v} \dot{\phi}_{s+v} (p_{s+a^*(s),s}^\dagger - q_{s+a^*(s)}^{B,s+v}) dv.$$

Because $p_{s+a^*(s),s}^\dagger = q_{s+a^*(s)}^{B,s}$ so $p_{s+a^*(s),s}^\dagger - q_{s+a^*(s)}^{B,s+v} = q_{s+a^*(s)}^{B,s} - q_{s+a^*(s)}^{B,s+v} \geq 0$ for all $v \in [0, a^*(s)]$ with equality only at $v = 0$. Moreover, $\dot{\phi}_t = -(\lambda_0 + \lambda_1)(p_0 - p^*)e^{-(\lambda_0 + \lambda_1)t}$ for all t , so $\dot{\phi}_t > 0$ if and only if $p_0 < p^*$, and $\dot{\phi}_{s+a^*(s)}$ and $\dot{\phi}_{s+v}$ have the same sign. Thus, the numerator is positive if and only if $p_0 < p^*$.

Putting it all together, $\frac{da^*(s)}{ds}$ has the same sign as $\frac{\partial p_{t,s}^\dagger}{\partial t}|_{t=s+a^*(s)}$ which is equal to $\frac{dp_{s+a,s}^\dagger}{ds}|_{a=a^*(s)}$. The above argument shows $\frac{dp_{s+a,s}^\dagger}{ds}|_{a=a^*(s)}$ is positive if and only if $p_0 < p^*$. This proves (ii).

For (iii), first suppose $p_0 = p^*$. Then $\phi_t = p^*$ for every t , so

$$p_{s+a,s}^\dagger = \frac{e^{-\mu a} p^* + (1 - p^*) \int_0^a \mu e^{-\mu v} q_{s+a}^{B,s+v} dv}{e^{-\mu a} + (1 - p^*) \int_0^a \mu e^{-\mu v} dv}.$$

Because $q_{s+a}^{B,s+v} = p^*(1 - e^{-(\lambda_0 + \lambda_1)(a-v)})$, the right-hand side is independent of s . Moreover, $q_{s+a}^{B,s} = p^*(1 - e^{-(\lambda_0 + \lambda_1)a})$ is also independent of s . Hence, when $p_0 = p^*$, $H(s, a)$ is independent of s . To emphasize this independence, define $H^*(a) := H(s, a)|_{p_0=p^*}$.

By Lemma 12, H^* has a unique zero $\alpha^* \in (0, \infty)$ and crosses zero from negative to positive. Therefore, $H^*(a) < 0$ for $a < \alpha^*$ and $H^*(a) > 0$ for $a > \alpha^*$.

We next show that, for every fixed $a > 0$, $\lim_{s \rightarrow \infty} H(s, a) = H^*(a)$. Recall that $\phi_t = p^* + (p_0 - p^*)e^{-(\lambda_0 + \lambda_1)t}$. Consequently, uniformly over $v \in [0, a]$,

$$|\phi_{s+v} - p^*| \leq |p_0 - p^*|e^{-(\lambda_0 + \lambda_1)s} \rightarrow 0.$$

Because $0 \leq q_{s+a}^{B,s+v} \leq p^*$, it follows that

$$\sup_{v \in [0, a]} |\mu e^{-\mu v} (1 - \phi_{s+v}) q_{s+a}^{B,s+v} - \mu e^{-\mu v} (1 - p^*) q_{s+a}^{B,s+v}| \leq \mu p^* |p_0 - p^*| e^{-(\lambda_0 + \lambda_1)s} \rightarrow 0.$$

Thus,

$$\int_0^a \mu e^{-\mu v} (1 - \phi_{s+v}) q_{s+a}^{B,s+v} dv \rightarrow \int_0^a \mu e^{-\mu v} (1 - p^*) q_{s+a}^{B,s+v} dv.$$

Similarly,

$$\int_0^a \mu e^{-\mu v} (1 - \phi_{s+v}) dv \rightarrow \int_0^a \mu e^{-\mu v} (1 - p^*) dv.$$

We also have $\phi_{s+a} \rightarrow p^*$. Because $e^{-\mu a} + (1 - p^*) \int_0^a \mu e^{-\mu v} dv > 0$, (15) implies $p_{s+a,s}^\dagger \rightarrow p_{s+a,s}^\dagger|_{p_0=p^*}$. Since $q_{s+a}^{B,s}$ is independent of both s and p_0 , $H(s, a) \rightarrow H^*(a)$ for a fixed $a > 0$.

Now fix any $\varepsilon \in (0, \alpha^*)$. Because H^* crosses the unique α^* from negative to positive $H^*(\alpha^* - \varepsilon) < 0 < H^*(\alpha^* + \varepsilon)$. Applying the convergence above at the two fixed arguments $\alpha^* - \varepsilon$ and $\alpha^* + \varepsilon$, there exists $S_\varepsilon < \infty$ such that, for every $s \geq S_\varepsilon$, $H(s, \alpha^* - \varepsilon) < 0 < H(s, \alpha^* + \varepsilon)$. By Lemma 12, $a^*(s)$ is the unique zero of $H(s, \cdot)$, and $H(s, \cdot)$ crosses from negative to positive at this zero. Therefore, for every $s \geq S_\varepsilon$, $\alpha^* - \varepsilon < a^*(s) < \alpha^* + \varepsilon$. Because this holds for every $\varepsilon \in (0, \alpha^*)$, it follows that $\lim_{s \rightarrow \infty} a^*(s) = \alpha^*$.

A.3 Proof of Corollary 1

By Proposition 1(i), $s + a^*(s)$ is increasing in s , which implies that for each $t \geq a^*(0)$, there is a unique $\tau(t) \geq 0$ satisfying $\tau(t) + a^*(\tau(t)) = t$. Moreover $\tau(t)$ is increasing in t . The equilibrium condition at each t says

$$p_t = p_{t, \tau(t)}^\dagger = q_t^{B, \tau(t)} = p^* (1 - e^{-(\lambda_0 + \lambda_1)(t - \tau(t))}) = p^* (1 - e^{-(\lambda_0 + \lambda_1)a^*(\tau(t))}).$$

Because $\tau(t)$ and the right-hand side are increasing in their respective arguments, p_t inherits the monotonicity of $a^*(s)$. Proposition 1 (ii) then implies that $p_{t,s}^\dagger$ is strictly increasing when $p_0 < p^*$, strictly decreasing when $p_0 > p^*$, and constant when $p_0 = p^*$.

Finally, as $t \rightarrow \infty$, $\tau(t) \rightarrow \infty$. By Proposition 1 (iii), $a^*(s) \rightarrow \alpha^*$, $\lim_{t \rightarrow \infty} p_{t,s}^\dagger = p^* (1 - e^{-(\lambda_0 + \lambda_1)\alpha^*}) < p^*$, where the inequality follows because α^* is finite.

A.4 Proof of Proposition 2

For each $t \geq a^*(0)$, define $\tau(t)$ implicitly by $\tau(t) + a^*(\tau(t)) = t$; that is, $\tau(t)$ is the most recent timestamp for bad evidence that is disclosed by date t . For $t < a^*(0)$, set $\tau(t) = -\infty$.

Now fix any $t \geq 0$. Let ν_s^G and ν_s^B denote the probability in equilibrium that good

evidence and bad evidence, respectively, obtained at time s is disclosed by time t . Fix an arbitrary alternative disclosure strategy and let $\tilde{\nu}_s^G$ and $\tilde{\nu}_s^B$ denote the analogous probabilities. Note that $\nu_s^G = 1$ for all $s \in [0, t]$, and $\nu_s^B = 1$ for all $s \leq \tau(t)$ and $\nu_s^B = 0$ for all $s > \tau(t)$. Let p_t and $\tilde{p}_t$ denote the beliefs conditional on no disclosure in the equilibrium and under the alternative disclosure strategy, respectively. Finally, let N_t and $\tilde{N}_t$ denote the probability of no disclosure under the equilibrium and alternative strategies, respectively.

By construction, we have (i) $p_t \leq q_t^{G,s}$ for all $s \in [0, t]$, (ii) $p_t \leq q_t^{B,s}$ for $s \in [0, \tau(t)]$, and (iii) $q_t^{B,s} \leq p_t$ for $s \in (\tau(t), t]$.

By the law of iterated expectations,

$$\phi_t = N_t p_t + \int_0^t \phi_s \mu e^{-\mu s} \nu_s^G q_t^{G,s} ds + \int_0^t (1 - \phi_s) \mu e^{-\mu s} \nu_s^B q_t^{B,s} ds \quad (16)$$

and also

$$\phi_t = \tilde{N}_t \tilde{p}_t + \int_0^t \phi_s \mu e^{-\mu s} \tilde{\nu}_s^G q_t^{G,s} ds + \int_0^t (1 - \phi_s) \mu e^{-\mu s} \tilde{\nu}_s^B q_t^{B,s} ds, \quad (17)$$

where the belief at time t after disclosure (at any time) of good or bad evidence with timestamp s , namely $q_t^{G,s}$ or $q_t^{B,s}$, is independent of the conjectured strategy. Equating the right hand sides of (16) and (17) and subtracting p_t from both yields

$$\begin{aligned} & 0 + \int_0^t \phi_s \mu e^{-\mu s} \nu_s^G (q_t^{G,s} - p_t) ds + \int_0^t (1 - \phi_s) \mu e^{-\mu s} \nu_s^B (q_t^{B,s} - p_t) ds \\ &= \tilde{N}_t (\tilde{p}_t - p_t) + \int_0^t \phi_s \mu e^{-\mu s} \tilde{\nu}_s^G (q_t^{G,s} - p_t) ds + \int_0^t (1 - \phi_s) \mu e^{-\mu s} \tilde{\nu}_s^B (q_t^{B,s} - p_t) ds \\ &\iff \tilde{N}_t (\tilde{p}_t - p_t) = \int_0^t \phi_s \mu e^{-\mu s} (\nu_s^G - \tilde{\nu}_s^G) (q_t^{G,s} - p_t) ds \\ &\quad + \int_0^t (1 - \phi_s) \mu e^{-\mu s} (\nu_s^B - \tilde{\nu}_s^B) (q_t^{B,s} - p_t) ds. \end{aligned} \quad (18)$$

Now for all $s \in [0, t]$, we have $1 = \nu_s^G \geq \tilde{\nu}_s^G$ and $q_t^{G,s} - p_t \geq 0$, so the first integral on the right hand side of (18) is nonnegative. We then turn to the integrand of the second integral. For $s \in [0, \tau(t)]$, we have $1 = \nu_s^B \geq \tilde{\nu}_s^B$ and $q_t^{B,s} \geq p_t$, so the integrand is nonnegative. And for $s \in (\tau(t), t]$, we have $0 = \nu_s^B \leq \tilde{\nu}_s^B$ and $q_t^{B,s} \leq p_t$, so again the integrand is nonnegative. Hence, the right hand side of (18) is nonnegative. As $\tilde{N}_t \geq e^{-\mu t} > 0$, we must have $\tilde{p}_t \geq p_t$.

A.5 Proof of Proposition 3

Suppose there exists such an equilibrium, and suppose the sender obtains good evidence at time 0. Let $F(T)$ denote the sender's discounted payoff from disclosing at time T . By disclosing immediately, the sender's payoff is $F(0) = \int_0^\infty e^{-rt} [p^* + (1 - p^*)e^{-(\lambda_0 + \lambda_1)t}] dt = \frac{\lambda_0 + r}{r(\lambda_0 + \lambda_1 + r)}$. By instead disclosing at some time $\varepsilon > 0$, the sender's payoff is $F(\varepsilon) = \int_0^\varepsilon e^{-rt} p_t dt + \int_\varepsilon^\infty e^{-rt} [p^* + (1 - p^*)e^{-(\lambda_0 + \lambda_1)(t - \varepsilon)}] dt = \int_0^\varepsilon e^{-rt} p_t dt + e^{-r\varepsilon} F(0)$. If $p_0 > \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}$, then since p_t is right-continuous, we have for sufficiently small $\varepsilon > 0$ that

$$\int_0^\varepsilon e^{-rt} p_t dt > \int_0^\varepsilon e^{-rt} \left(\frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r} \right) dt = \frac{1 - e^{-r\varepsilon}}{r} \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r} = (1 - e^{-r\varepsilon}) F(0).$$

Hence, $F(\varepsilon) > (1 - e^{-r\varepsilon})F(0) + e^{-r\varepsilon}F(0) = F(0)$. Thus, immediate disclosure is not optimal at time 0.

A.6 Proof of Lemma 1

Consider a strategy with t_0 as in the lemma. Then $G_{0,t} > 0$ and $G_{1,t} > 0$, where $G_{i,t}$ is defined in Section 4.2. It follows that after disclosure of good evidence at time t_0 , the posterior is $q_{t_0} = \frac{G_{1,t}}{G_{0,t} + G_{1,t}} < 1$, and the payoff of such disclosure is $\Pi(q_{t_0}) := \frac{p^*}{r} + \frac{q_{t_0} - p^*}{\lambda_0 + \lambda_1}$. By disclosing at time $t_0 + \epsilon$, the receiver's belief is at least $\underline{q}(\epsilon) := e^{-(\lambda_0 + \lambda_1)\epsilon}$, since the maximum age of the evidence is ϵ , and the continuation payoff from t_0 of doing so is at least $e^{-r\epsilon}\Pi(\underline{q}(\epsilon))$, which tends to $\Pi(1) > \Pi(q_{t_0})$ as $\epsilon \rightarrow 0$.

A.7 Proof of Theorem 2

A.7.1 Setup and Laws of Motion

We use the notation and laws of motion introduced in Section 4.2. In particular, G_i denotes the joint probability that there is undisclosed good evidence and the time- t state is i ; $S_t = G_{0,t} + G_{1,t}$ denotes the total stock of undisclosed good evidence; $q_t = G_{1,t}/S_t$ is the receiver's belief after disclosure of good evidence; and p_t is the no-disclosure belief. During a stockpiling phase, $\kappa_t = 0$.

During either the stockpiling or purging phase, (G_1, G_0, p) have the following laws of motion (5)-(7). We use the change of variables $S = G_0 + G_1$ and $q = G_1/S$ and differentiate

to obtain

$$\dot{S}_t = (p_t - S_t q_t)\mu - S_t \kappa_t(1 - S_t) \quad (19)$$

$$\dot{q}_t = -\lambda_1 q_t + (1 - q_t)\mu \frac{p_t - S_t q_t}{S_t} \quad (20)$$

During the transparency phase, we have $q_t \equiv 1$ and p evolves according to

$$\dot{p}_t = -\lambda_1 p_t - \mu p_t(1 - p_t). \quad (21)$$

Note that p is decreasing during the transparency phase.

A.7.2 Local incentive compatibility

We will make heavy use of local incentive compatibility. Suppose the sender possesses good evidence at time t_0 and plans to disclose it at time $t \geq t_0$. Then for $s \geq t$, we have $p_s = q_t e^{-\lambda_1(s-t)}$. The sender's payoff is

$$F(t; t_0) := \int_{t_0}^t e^{-r(s-t_0)} p_s ds + \int_t^\infty e^{-r(s-t_0)} q_t e^{-\lambda_1(s-t)} ds.$$

The sender is indifferent disclosing at time t_0 and time $t_0 + \epsilon$ for all sufficiently small ϵ if and only if $F_t(t; t_0)|_{t=t_0} = 0$; equivalently,

$$0 = p_{t_0} - q_{t_0} + \int_{t_0}^\infty (\lambda_1 q_{t_0} + \dot{q}_{t_0}) e^{-r(s-t_0)} e^{-\lambda_1(s-t_0)} ds = p_{t_0} - q_{t_0} + (\lambda_1 q_{t_0} + \dot{q}_{t_0}) \frac{1}{\lambda_1 + r}.$$

Rearranging and replacing t_0 with t yields the local indifference condition

$$\dot{q}_t = r q_t - (\lambda_1 + r) p_t. \quad (22)$$

If $\dot{q}_t > r q_t - (\lambda_1 + r) p_t$, then for small enough ϵ , the sender strictly prefers to disclose at $t + \epsilon$ rather than at time t . If $\dot{q}_t < r q_t - (\lambda_1 + r) p_t$, then the sender strictly prefers to disclose at time t than at time $t + \epsilon$ for all sufficiently small ϵ .

A.7.3 Identifying $\underline{p}$ and $\bar{p}$

In this section we prove the following lemma.

Lemma 13. *Any three-phase equilibrium must start in the stockpiling phase if $p_0 > \bar{p}$, the transparency phase if $p_0 \leq \underline{p}$, and the purging phase if $p_0 \in (\underline{p}, \bar{p}]$, where $\bar{p} := \frac{r+\lambda_1/2}{\lambda_1+r}$ and $\underline{p} := \frac{r}{\lambda_1+r}$.*

Proof. By definition, any three-phase equilibrium must start in one of the three phases. We consider each possibility separately and derive necessary conditions on p_0 . We then conclude the proof by observing that the conditions are mutually exclusive, so that p_0 determines the starting phase as in the proposition.

Starting in stockpiling phase: We argue that if the game starts in the stockpiling phase, then a sender who obtains good evidence at time 0 must be willing to wait an instant, and this implies $p_0 > \bar{p} = \frac{r + \frac{\lambda_1}{2}}{\lambda_1 + r}$.

A necessary and sufficient condition to start in the stockpiling phase is that for some time $\underline{T} > 0$, waiting until $\underline{T}$ to disclose must be weakly preferred to disclosing at time t for a sender who obtains good evidence at time $t \in [0, \underline{T}]$. Hence, recalling (22), a necessary condition for starting in stockpiling is that for some $\underline{T} > 0$,

$$\dot{q}_{\underline{T}} \geq r q_{\underline{T}} - (\lambda_1 + r) p_{\underline{T}}. \quad (23)$$

Using $\kappa_t \equiv 0$, the laws of motion for beliefs admit a (unique) closed-form solution:

$$G_{1,t} = e^{-\lambda_1 t} p_0 (1 - e^{-\mu t}) \quad (24)$$

$$G_{0,t} = e^{-\lambda_1 t} p_0 \left(\frac{\lambda_1 e^{-t\mu} + \mu e^{\lambda_1 t}}{\mu + \lambda_1} - 1 \right) \quad (25)$$

$$p_t = e^{-\lambda_1 t} p_0 \quad (26)$$

$$q_t = \frac{G_{1,t}}{G_{0,t} + G_{1,t}} = \frac{(e^{\mu t} - 1)(\lambda_1 + \mu)}{(e^{(\lambda_1 + \mu)t} - 1)\mu} \text{ for } t > 0. \quad (27)$$

Recall that $q_0 = 1$; it is easy to calculate the right derivative $\dot{q}_0 = -\frac{\lambda_1}{2}$.

Using (26), we get the necessary condition

$$p_0 \geq b(t) := \frac{e^{\lambda_1 t} (r q_t - \dot{q}_t)}{\lambda_1 + r}$$

for some $t > 0$, where q_t is given by (27). Define $b(0) := \lim_{t \downarrow 0} b(t) = \frac{r \cdot q_0 - \dot{q}_0}{\lambda_1 + r} = \frac{r \cdot 1 + \frac{\lambda_1}{2}}{\lambda_1 + r} = \bar{p}$. We claim that $b'(t) > 0$ for all $t > 0$. Direct calculation shows that

$$\begin{aligned} b'(t) &= \frac{e^{\lambda_1 t} (\lambda_1 + \mu)}{(e^{(\lambda_1 + \mu)t} - 1)^3 (\lambda_1 + r) \mu} P(t), \quad \text{where} \\ P(t) &:= \mu(\lambda_1 + \mu + r) e^{t(2\lambda_1 + 2\mu)} - (\lambda_1 + \mu)(2\lambda_1 + \mu + r) e^{t(\lambda_1 + 2\mu)} \\ &\quad + (2\lambda_1^2 + 3\lambda_1 \mu + \lambda_1 r + \mu^2 - \mu r) e^{t(\lambda_1 + \mu)} - (\lambda_1 + \mu)(\mu - r) e^{t\mu} - \lambda_1 r. \end{aligned}$$

For each of two cases $\mu \geq r$ and $\mu < r$, the exponential polynomial $P(t)$ has three sign changes and thus at most three real roots counting multiplicity. It is also easy to check that there is a triple root at $t = 0$. Hence, there are no other real roots. Also, since the leading term is positive, $P(t)$ is positive for sufficiently large t . We conclude that for all $t > 0$, we have $P(t) > 0$ and therefore $b'(t) > 0$. It follows that $p_0 \geq b(t)$ for some $t > 0$ only if

$$p_0 > b(0) = \bar{p}.$$

This is exactly the inequality asserted in Lemma 13.

For later use, let us define $c(t) = \dot{q}_t - rq_t + (\lambda_1 + r)p_t$. Assume $p_0 > \bar{p}$. We show that c crosses 0 from above at the unique point t where $p_0 = b(t)$. We have

$$\begin{aligned} c(t) &= (\lambda_1 + r)e^{-\lambda_1 t}(p_0 - b(t)) \\ \implies c'(t) &= -\lambda_1 c(t) - b'(t)(\lambda_1 + r)e^{-\lambda_1 t}. \end{aligned}$$

When $c(t) = 0$, the right hand side has the same sign as $-b'(t) < 0$, as desired.

Starting in transparency phase: In this case $q_0 \equiv 1$ and $\dot{q}_0 \equiv 0$, so at time 0 the left hand side of (22) is 0 and the right hand side is $r - (\lambda_1 + r)p_0$. Now for immediate disclosure to be optimal, we must have $0 \leq r - (\lambda_1 + r)p_0$, or equivalently, $p_0 \leq \frac{r}{\lambda_1 + r} = \underline{p}$.

Starting in purging phase: The sender must be indifferent between disclosing good evidence at time 0 and disclosing at time t for all sufficiently small $t > 0$, the local IC constraint (22) must hold for all $t \in [0, \epsilon)$ for sufficiently small $\epsilon > 0$.

At $t = 0$, we have $q_0 = 1$ and $\dot{q}_0 \leq 0$ as q is bounded above by 1. This implies $p_0 \geq \frac{r}{\lambda_1 + r}$. Moreover, we can rule out $p_0 = \frac{r}{\lambda_1 + r}$ as follows. Note that this would require $\dot{q}_0 = 0$. Since q is bounded above by 1, we have $\ddot{q}_0 \leq 0$. If indifference holds for all $t \in [0, \epsilon)$, then $\ddot{q}_0 = r\dot{q}_0 - (\lambda_1 + r)\dot{p}_0 = 0 - (\lambda_1 + r)\dot{p}_0 \geq (\lambda_1 + r)\lambda_1 p_0 > 0$, where the bound $\dot{p}_0 \leq -\lambda_1 p_0$ follows from $p_t \leq \phi_t = \phi_0 e^{-\lambda_1 t}$. This contradicts $\ddot{q}_0 \leq 0$, so we conclude that the game can begin with a purging phase only if $p_0 > \frac{r}{\lambda_1 + r}$.

We argue that $\dot{q}_0 \geq -\frac{\lambda_1}{2}$. Let $X_t = t - \tau$ be the (possibly random) age of evidence disclosed at t . Then

$$\dot{q}_0 = -\lambda_1 \lim_{t \rightarrow 0} \frac{\mathbb{E}[X_t]}{t}.$$

Expanding and writing $\gamma_s := \phi_s \mu e^{-\mu s}$ yields

$$\lim_{t \rightarrow 0} \frac{\mathbb{E}[X_t]}{t} = \lim_{t \rightarrow 0} \frac{\int_0^t (t-s) \gamma_s e^{-\int_s^t \kappa_u du} ds}{t \int_0^t \gamma_s e^{-\int_s^t \kappa_u du} ds} = \lim_{t \rightarrow 0} \frac{\int_0^t (t-s) e^{-\int_s^t \kappa_u du} ds}{t \int_0^t e^{-\int_s^t \kappa_u du} ds},$$

where we have used that γ is continuous. Since the density $\frac{e^{-\int_s^t \kappa_u du}}{\int_0^t e^{-\int_x^t \kappa_u du} dx}$ is nondecreasing in s , while the function $t-s$ is decreasing, the weighted average is bounded above by replacing the density with a uniform density. Thus,

$$\lim_{t \rightarrow 0} \frac{\mathbb{E}[X_t]}{t} \leq \lim_{t \rightarrow 0} \frac{\int_0^t (t-s) ds}{t \int_0^t ds} = \lim_{t \rightarrow 0} \frac{\frac{t^2}{2}}{t^2} = \frac{1}{2}.$$

We conclude that $\dot{q}_0 \geq -\frac{\lambda_1}{2}$, and thus

$$p_0 = \frac{r - \dot{q}_0}{\lambda_1 + r} \leq \frac{r + \frac{\lambda_1}{2}}{\lambda_1 + r}.$$

Summary We have shown that starting in the stockpiling phase requires $p_0 > \bar{p}$, the purging phase $p_0 \in (\underline{p}, \bar{p}]$, and the transparency phase $p_0 \leq \underline{p}$. Since a three-phase equilibrium must start in one of these three phases, the only possibility is stockpiling for $p_0 > \bar{p}$, transparency for $p_0 \leq \underline{p}$, and purging for $p_0 \in (\underline{p}, \bar{p}]$. $\square$

We now establish existence and uniqueness within the class of three-phase equilibria for p_0 in each of the three intervals.

A.7.4 Low p_0 .

Suppose $p_0 \leq \frac{r}{\lambda_1 + r}$ and that the receiver conjectures that bad evidence is never disclosed, while good evidence is disclosed immediately.

Then $q_t \equiv 1$ and p evolves according to (21). Note that because p is strictly decreasing, if $p_0 \leq \underline{p}$ then this inequality remains true at all future times. Hence, immediate disclosure of evidence is optimal at all histories, including after off-path delay of disclosure.

A.7.5 High p_0 .

Suppose that $p_0 > \frac{r + \lambda_1/2}{r + \lambda_1}$. Fix any candidate three-phase equilibrium. Since it must start in the stockpiling phase, the phases are characterized by cutoff times $0 < \underline{T}_0 < \bar{T}_0 < \infty$. Let $(\kappa_t)_{t \in [\underline{T}_0, \bar{T}_0]}$ be the disclosure intensity during the purging phase. We derive necessary conditions that uniquely determine these objects and then verify that the sender's strategy is optimal.

Let (p, q, S) denote the belief/stock variables in the candidate equilibrium. It is useful to define auxiliary variables (p^0, S^0, q^0) as the belief/stock variables under an alternative conjecture that good evidence and bad evidence are never disclosed (with q^0 determined by neutral belief updating).

Since both conjectures prescribe no disclosure on $[0, \underline{T}_0)$, and the variables (p_t, q_t, S_t) and (p_t^0, q_t^0, S_t^0) are continuous at $\underline{T}_0$ (since there are no disclosure atoms at $\underline{T}_0$), we have

$$(p_t, q_t, S_t) = (p_t^0, q_t^0, S_t^0) \quad \text{for all } t \leq \underline{T}_0.$$

These have closed form expressions given by (24)-(27) where $q_0 = 1$ and $q_t = G_{1,t}/(G_{0,t} + G_{1,t})$ for $t > 0$.

Lemma 14. *We have $\underline{T}_0 = \underline{T}$, where $\underline{T}$ is the unique time at which local IC (22) holds under the no-disclosure beliefs:*

$$\dot{q}_t^0 = r q_t^0 - (\lambda_1 + r) p_t^0.$$

Proof. In the purging phase, q must satisfy local indifference,

$$\dot{q}_t = r q_t - (\lambda_1 + r) p_t \quad \text{for } t \in [\underline{T}_0, \bar{T}_0), \quad (28)$$

where we use the right-derivative. Also, by (20) at $t = \underline{T}_0$,

$$\dot{q}_{\underline{T}_0} = -\lambda_1 q_{\underline{T}_0} + (1 - q_{\underline{T}_0}) \mu \frac{p_{\underline{T}_0} - S_{\underline{T}_0} q_{\underline{T}_0}}{S_{\underline{T}_0}}, \quad (29)$$

$$\dot{q}_{\underline{T}_0}^0 = -\lambda_1 q_{\underline{T}_0}^0 + (1 - q_{\underline{T}_0}^0) \mu \frac{p_{\underline{T}_0}^0 - S_{\underline{T}_0}^0 q_{\underline{T}_0}^0}{S_{\underline{T}_0}^0}. \quad (30)$$

Since $(p_{\underline{T}_0}, S_{\underline{T}_0}, q_{\underline{T}_0}) = (p_{\underline{T}_0}^0, S_{\underline{T}_0}^0, q_{\underline{T}_0}^0)$, (29) and (30) imply $\dot{q}_{\underline{T}_0} = \dot{q}_{\underline{T}_0}^0$. Combining this with (28) and again using $(p_{\underline{T}_0}, S_{\underline{T}_0}, q_{\underline{T}_0}) = (p_{\underline{T}_0}^0, S_{\underline{T}_0}^0, q_{\underline{T}_0}^0)$ gives

$$\dot{q}_{\underline{T}_0}^0 = r q_{\underline{T}_0}^0 - (\lambda_1 + r) p_{\underline{T}_0}^0. \quad (31)$$

Recall that from the construction of $\bar{p}$ earlier, for $p_0 > \bar{p} = b(0)$, there is a unique value of $\underline{T}_0 > 0$ at which $p_0 = b(\underline{T}_0)$, which is equivalent to (31). We denote this unique value by $\underline{T}$. Hence, in any three-phase equilibrium, $\underline{T}_0 = \underline{T}$; in other words, the stockpiling phase must have length $\underline{T}$. $\square$

We now characterize the purging phase solution: κ_t and $\bar{T}$. The initial values for $(p_{\underline{T}}, G_{1,\underline{T}}, G_{0,\underline{T}}, q)$ are already given from the end of the stockpiling phase. In the purg-

ing phase, (p, G_1, G_0, q, κ) must satisfy the local indifference condition (22) and the laws of motion (5)-(7). In addition, q must also satisfy, by definition, $q_t = G_{1,t}/(G_{1,t} + G_{0,t})$ at each t in $[\underline{T}_0, \bar{T}_0]$.¹⁵

This system first reduces to the system in (p, q, S, κ) given by (7), (19), (20), and (22). In the next two steps, we eliminate S_t and κ_t .

Next, we solve for S_t by using the right side of (22) to replace $\dot{q}_t$ in (20):

$$\begin{aligned} \dot{q}_t &= -\lambda_1 q_t + (1 - q_t) \mu \frac{p_t - S_t q_t}{S_t} = r q_t - (\lambda_1 + r) p_t \\ \implies S_t &= S(p_t, q_t) := \frac{\mu p_t (1 - q_t)}{q_t [\lambda_1 + r + \mu(1 - q_t)] - (\lambda_1 + r) p_t}. \end{aligned} \quad (32)$$

A sufficient condition for $S(p_t, q_t)$ to be well defined is that $p_t < q_t \leq 1$, which we will verify.

To solve for κ_t , differentiate (32), substitute in the expressions for $\dot{p}$ and $\dot{q}$ from (7) and (22), and equate the resulting expression to the right hand side of (19). Solving for κ_t yields

$$\kappa_t = \kappa(p_t, q_t) := \frac{A_t(q_t[\lambda_1 + r + \mu(1 - q_t)] - (\lambda_1 + r)p_t)}{(q_t - p_t)(1 - q_t)[\lambda_1 + r + \mu(1 - q_t)]}, \quad \text{where} \quad (33)$$

$$A_t := A(p_t, q_t) := r + \lambda_1 + \mu + (r - \mu)q_t - 2p_t(\lambda_1 + r). \quad (34)$$

As long as $0 < p_t < q_t < 1$, which we will verify in our solution, the denominator will be positive, so $\kappa(p_t, q_t)$ will be well defined. We will also need to verify that $\kappa_t \geq 0$ at all times for it to be feasible as an arrival rate. Since $0 < p_t < q_t < 1$ implies the second factor in the numerator is positive, it will be sufficient to show that $A_t \geq 0$ for all $t \in [\underline{T}, \bar{T}]$.

After eliminating S_t and κ_t in this way, we obtain a reduced IVP consisting of the ODEs

$$\dot{p}_t = p_t \frac{2(\lambda_1 + r)\mu p_t + \mu q_t(\lambda_1 + \mu - r) - (\lambda_1 + \mu)(\lambda_1 + r + \mu)}{\lambda_1 + r + \mu(1 - q_t)} \quad (35)$$

$$\dot{q}_t = r q_t - (\lambda_1 + r) p_t, \quad (36)$$

with initial conditions $(p_{\underline{T}}, q_{\underline{T}})$ from the end of the stockpiling phase.

Note that the right hand sides of (35) and (36) are time-invariant, C^1 functions of (p_t, q_t) on the domain $U := \{(p, q) \in \mathbb{R}^2 : q < 1 + \frac{\lambda_1 + r}{\mu}\}$. In particular, they are locally Lipschitz continuous, uniformly in time. By the Picard-Lindelöf theorem, there exists a unique solution to the IVP with initial conditions on some interval $[\underline{T}, \underline{T} + \varepsilon]$ with $\varepsilon > 0$. Let $[\underline{T}, T^{\max})$ denote the maximal interval of existence (and uniqueness), where $T^{\max}$ is possibly ∞ .

¹⁵Since $\underline{T}_0 > 0$, $G_{1,\underline{T}_0}/(G_{1,\underline{T}_0} + G_{0,\underline{T}_0})$ is well defined.

Define $R \subset U$ by

$$R := \{(p, q) : 0 < p < q < 1 \text{ and } A(p, q) > 0\}.$$

Then on R , (i) κ_t is well defined via (33) and strictly positive, and (ii) S_t is well defined via (32). Define the (possibly infinite) time $T := \inf\{t \in [\underline{T}, T^{\max}) : (p_t, q_t) \notin R\}$.

Lemma 15. *We have that $(p_{\underline{T}}, q_{\underline{T}})$ lies in the interior of R . Hence, $T > \underline{T}$.*

Proof. In the stockpiling phase, we have

$$\dot{p}_t = -\lambda_1 p_t \tag{37}$$

$$\dot{q}_t = -\lambda_1 q_t + (1 - q_t)\mu \frac{p_t - S_t q_t}{S_t} \tag{38}$$

$$\dot{S}_t = (p_t - S_t q_t)\mu, \tag{39}$$

with initial conditions $(p_0, q_0, S_0) = (p_0, 1, 0)$. That p_t is strictly positive for all $t \in [0, \underline{T}]$ is standard by either the comparison theorem and is also evident from the closed form solution.

It is useful to show that $S_t > 0$ until $\underline{T}$. We have $\dot{S}_0 > 0$, so $S_t > 0$ for all sufficiently small t . Now write (39) as $\dot{S}_t = f(t, S_t)$, and define $y_t \equiv 0$. Then $S_t > y_t$ and $\dot{S}_t - f(t, S_t) = 0 > -p_t \mu = y_t - f(t, y_t)$. By the comparison theorem, $S_t > y_t = 0$ for all $t > 0$.

The comparison theorem also implies $q_t > 0$ for all $t \in [0, \underline{T}]$; moreover, $q_t < 1$ for all $t \in (0, \underline{T}]$, since $\dot{q}_0 = -\frac{\lambda_1}{2} < 0$ and the RHS of the q -ODE is negative whenever $q_t = 1$.

Also note that $N_t := p_t - S_t q_t = p_t - G_{1,t}$, which has ODE $\dot{N}_t = -(\lambda_1 + \mu)N_t$. Since $N_0 = p_0 - G_{1,0} = p_0 \in (0, 1)$, we have $N_t = p_t - G_{1,t} > 0$ for all $t \in [0, \underline{T}]$.

Define $\delta_t := q_t - p_t$, where $\delta_0 = 1 - p_0 > 0$. We have

$$\dot{\delta}_t = -\lambda_1 \delta_t + (1 - q_t)\mu \frac{N_t}{S_t}. \tag{40}$$

The inequalities just established imply the second term is positive for all $t \in (0, \underline{T}]$. Hence, by the comparison theorem, we have $\delta_t > 0$ for all $t \in [0, \underline{T}]$.

To prove that $A_{\underline{T}} > 0$, recall from earlier in the proof that $c'(\underline{T}) < 0$. Hence, $c(\underline{T})$ crosses 0 from above at $\underline{T}$. Now $c(\underline{T}) = 0$ implies

$$0 = \dot{q}_{\underline{T}} - r q_{\underline{T}} + (\lambda_1 + r) p_{\underline{T}} \tag{41}$$

, and combining this with (38) at $t = \underline{T}$ yields $S_{\underline{T}} = S(p_{\underline{T}}, q_{\underline{T}})$. Now when $c(t) = 0$, some algebra shows that $c'(t) = -\frac{(\lambda_1 + r)(q_t - p_t)}{1 - q_t} A_t$. Since $c(\underline{T}) = 0$ by construction and $c'(\underline{T}) < 0$, it follows that $A_{\underline{T}} > 0$. $\square$

The following lemma establishes that if the solution exits R at finite time T , it does so through $q_T = 1$.

Lemma 16. *If T is finite, then $q_T > p_T > 0$ and $A_T > 0$. Hence, $q_T = 1$.*

Proof. If $T < \infty$, then $T < T^{\max}$ by definition of T . Since the right hand side of (35) contains a factor of p_t , the comparison theorem implies $p_t > 0$ for all $t \in [\underline{T}, T^{\max})$, and in particular, $p_T > 0$.

Note that for all $t \in [\underline{T}, T)$, we have $(p_t, q_t) \in R$ and therefore κ_t is well-defined and strictly positive. Moreover, S_t is well-defined and strictly positive. Hence, by construction, the ODEs for (G_1, G_0, p) are all satisfied.

The original ODE for p can be written

$$\dot{p}_t = -\lambda_1 p_t - S_t \kappa_t (q_t - p_t). \quad (42)$$

Defining $\delta_t = q_t - p_t$ and subtracting this ODE for p from the indifference condition for q , we have (9). It follows that δ is strictly increasing on $[\underline{T}, T)$, so $\delta_T > 0$; in other words, $q_T > p_T$.

We now establish that $A_T > 0$. Since $A_t > 0$ for all $t < T$, we must have $A_T \geq 0$. Toward a contradiction, suppose $A_T = 0$. We claim that $\dot{A}_T > 0$. A_t is differentiated as

$$\begin{aligned} \dot{A}_t &= (r - \mu)\dot{q}_t - 2\dot{p}_t(\lambda_1 + r) \\ &= (r - \mu)[rq_t - (\lambda_1 + r)p_t] \\ &\quad - 2(\lambda_1 + r) \left[p_t \frac{2(\lambda_1 + r)\mu p_t + \mu q_t(\lambda_1 + \mu - r) - (\lambda_1 + \mu)(\lambda_1 + r + \mu)}{\lambda_1 + r + \mu(1 - q_t)} \right]. \end{aligned}$$

Now if $A_t = 0$, then $p_t = \frac{r + \lambda_1 + \mu + (r - \mu)q_t}{2(\lambda_1 + r)}$. Substituting this into the expression for $\dot{A}_t$ and simplifying yields

$$\begin{aligned} \dot{A}_t|_{A_t=0} &= C_0 + C_1 q_t, \quad \text{where} \\ C_0 &:= \frac{1}{2}(2\lambda_1 + \mu - r)(\lambda_1 + r + \mu) \\ C_1 &:= \frac{1}{2}(r - \mu)(2\lambda_1 + \mu + r). \end{aligned}$$

Since this is linear in q_t , it suffices to check the endpoints. Using $q_t = 1$ yields $C_0 + C_1 = \frac{1}{2}\lambda_1(3r + 2\lambda_1 + \mu) > 0$. To evaluate at a lower bound on q_t , recall that $A_T = 0$ implies

$$p_T = \frac{r + \lambda_1 + \mu + (r - \mu)q_T}{2(\lambda_1 + r)} < q_T,$$

as we have already shown $p_T < q_T$. This in turn yields $q_T > \underline{q} := \frac{\lambda_1 + \mu + r}{2\lambda_1 + \mu + r}$. Hence,

$$\begin{aligned} C_0 + C_1 \underline{q} &= \frac{1}{2}(2\lambda_1 + \mu - r)(\lambda_1 + r + \mu) + \frac{1}{2}(r - \mu)(\lambda_1 + \mu + r) \\ &= \lambda_1(\lambda_1 + r + \mu) > 0. \end{aligned}$$

As $q_t \in (\underline{q}, 1]$ it follows that $C_0 + C_1 q_t$ lies between two positive values obtained at the extreme values $\underline{q}$ and 1, and is therefore positive.

We conclude that $\dot{A}_T|_{A_T=0} > 0$, so A_t cannot intersect 0 from above at $t = T$. Thus, $A_T > 0$.

Having shown that $q_T > p_T > 0$ and $A_T > 0$, the only remaining possibility is $q_T = 1$. $\square$

We now show that T is finite, so Lemma 16 indeed applies.

If $T = \infty$, the only possibility is that $T^{\max} = \infty$; otherwise, the solution would have to exit every compact subset of U in finite time. Since $R \subset [0, 1]^2 \subset U$, this contradicts $T = \infty$. The following lemma rules out this possibility.

Lemma 17. *It does not hold that $T = T^{\max} = \infty$.*

Proof. Suppose $T = T^{\max} = \infty$. Then a unique solution exists for all time and remains in R . But the ODE for $\delta = q - p$ has solution

$$\delta_t = \delta_{\underline{T}} e^{\int_{\underline{T}}^t (r + S_s \kappa_s) ds} \geq \delta_{\underline{T}} e^{r(t - \underline{T})}$$

where we have used that $S_t > 0$ and $\kappa_t > 0$ for all t , since $(p_t, q_t) \in R$. In particular, δ reaches 1 in finite time. This contradicts the fact that in R , $0 < p_t < q_t < 1$. $\square$

To summarize, we have $T < \infty$, and the unique maximal solution exits R at time T with $q_T = 1$. We define $\bar{T} = T$. It follows that as $t \rightarrow \bar{T}$, $\kappa_t \rightarrow \infty$ and $S_t \rightarrow 0$.

Because q_t hits 1 from below at $\bar{T}$, we have $\dot{q}_{\bar{T}-} \geq 0$. In fact, we have $\dot{q}_{\bar{T}-} > 0$. To see this, suppose $\dot{q}_{\bar{T}-} = 0$, which implies $p_{\bar{T}} = \frac{r}{\lambda_1 + r}$. Evaluating the p -ODE at $(p_{\bar{T}}, q_{\bar{T}}) = (\frac{r}{\lambda_1 + r}, 1)$ yields $\dot{p}_{\bar{T}-} < 0$. But then using $q_{\bar{T}} = 1$ and $\dot{q}_{\bar{T}-} = 0$ yields $\ddot{q}_{\bar{T}-} = -(\lambda_1 + r)\dot{p}_{\bar{T}-} > 0$, which would imply $q_t > 1$ for sufficiently large $t < \bar{T}$, a contradiction. Given that $\dot{q}_{\bar{T}-} > 0$, we have $p_{\bar{T}} < \frac{r}{\lambda_1 + r}$.

As (22) holds throughout the purging phase and (p, q) are continuous at $\bar{T}$, it follows that $p_{\bar{T}} \leq \frac{r q_{\bar{T}}}{\lambda_1 + r} = \underline{p}$. From time $\bar{T}$ onward, we have $q_t = 1$, and optimality of the sender's strategy follows exactly as before.

A.7.6 Moderate p_0

Suppose $p_0 \in (\underline{p}, \bar{p}]$. In this case, the equilibrium must begin in the purging phase. For all $t \in (0, \bar{T})$, (p, q) must satisfy (35) and (36), and must obey initial conditions $(p_0, q_0) = (p_0, 1)$. Note that $A(p_0, 1) = 2r + \lambda_1 - 2p_0(\lambda_1 + r) \geq 0$ because $p_0 \leq \bar{p}$. As we already showed, $\dot{A}_T|_{A_T=0} > 0$, so $A_t > 0$ for all $t \in (0, \bar{T})$; moreover, $A_0 > 0$ if and only if $p_0 < \bar{p}$. By standard arguments, there exists a unique local solution to the system in (p, q) . Since $q_0 = 1$, the initial condition is not in R , but $p_0 > \underline{p}$ implies $\dot{q}_0 < 0$, so we have $0 < p_t < q_t < 1$ and $A_t > 0$ for all sufficiently small t ; in other words, the solution immediately enters R . A repeat of the arguments from the high- p_0 case establishes that the solution exits R at some finite time $\bar{T}$ through $q_{\bar{T}} = 1$. $S_t = S(p_t, q_t)$ and $\kappa_t = \kappa(p_t, q_t)$ are then well defined and strictly positive by (32) and (33) for all $t \in (0, \bar{T})$. We have $S_0 = S(p_0, 1) = 0$, while $\kappa(p_0, 1)$ is undefined; if $p_0 < \bar{p}$, we have $\lim_{t \rightarrow 0} \kappa(p_t, q_t) = +\infty$ and $\lim_{t \rightarrow 0} \kappa(p_t, q_t)$ is finite and positive otherwise. The variables G_0 and G_1 are recovered by $G_{1,t} = S_t q_t$ and $G_{0,t} = S_t - G_{1,t}$; note that their initial conditions $(G_{1,0}, G_{0,0}) = (0, 0)$ are satisfied. By construction, (5)-(7), (19), and (20) are satisfied at all $t > 0$ by construction.

A.8 Proofs for Section 4.3

We begin with preliminary definitions. We define q_t^B as the belief immediately after disclosure of bad evidence. We define $q_t^{B,s}$ as the belief at time t conditional on bad evidence arriving at time s . Let σ^G and σ^B denote the time at which good or bad evidence, respectively, is disclosed (with the restriction that at most one of these is finite (and the other $+\infty$), since only one piece of evidence can arrive); let $\sigma = \min\{\sigma^G, \sigma^B\}$.

We define the following stock variables conditional on no disclosure up to time t :

$$\begin{aligned} B_{i,t} &= \Pr(\tau \leq t \text{ and } \theta_\tau = 0 \text{ and } \theta_t = i | \sigma > t) \quad \text{for } i \in \{0, 1\}, \\ B_t &= \Pr(\tau \leq t \text{ and } \theta_\tau = 0 | \sigma > t) = B_{0,t} + B_{1,t} \\ \bar{q}_t^B &= \begin{cases} \frac{B_{1,t}}{B_t} & \text{if } B_t > 0 \\ 0 & \text{if } B_t = 0 \end{cases}. \end{aligned}$$

It is also sometimes convenient to work with the unnormalized (i.e., not dividing by the

probability of nondisclosure) stock variables. We define

$$\begin{aligned} B_{i,t}^{un} &= \Pr(\tau \leq t \text{ and } \theta_\tau = 0 \text{ and } \theta_t = i \text{ and } \sigma > t) \\ B_t^{un} &= \Pr(\tau \leq t \text{ and } \theta_\tau = 0 \text{ and } \sigma > t) \\ N_t^{un} &= \Pr(\tau > t). \end{aligned}$$

Before proving Proposition 5, we state and prove the following supporting lemma, which provides a useful lower bound on the evolution of p_t . Intuitively, under age-independent bad evidence disclosure, such disclosure can only lower the belief, so nondisclosure of bad evidence can only raise it. The only remaining sources of downward pressure on the nondisclosure belief are therefore (i) Markov switching from state 1 to 0, and (ii) the fact that under immediate good evidence disclosure, no news is bad news. Since the proof is unrelated to the sender's optimization, we defer it to the Online Appendix.

Lemma 18. *Suppose that good evidence is disclosed immediately. Then under any age-independent bad evidence disclosure strategy, for all t, s we have $p_{t+s} \geq p_t - (\lambda_1 + \mu)s$. By implication, $p_t \geq p_{t-}$.*

Proof of Proposition 5. For the first part of the proposition, suppose by way of contradiction that there exists some fixed good evidence strategy, with a positive probability of delay, such that this strategy can be part of an equilibrium for arbitrarily large r . Let t_0 be the infimum of all good evidence types that delay with positive probability. Then for any $\epsilon > 0$, there exists $t \in [t_0, t_0 + \epsilon)$ such that good evidence type t delays with positive probability. This means there exists some $\Delta_0 > 0$ such that there is positive probability of delay until at least $t + \Delta_0$. Moreover, there must exist $r_n \rightarrow \infty$ and an associated sequence of equilibria all of which contain this good evidence strategy, such that for each n , there exists $\Delta_n \geq \Delta_0$ such that delay until $t + \Delta_n$ is optimal. For each n , the payoff of disclosing at time t is

$$\Pi(q_{n,t}^G) := \frac{p^*}{r_n} + \frac{q_{n,t}^G - p^*}{\Lambda + r_n}, \quad (43)$$

and the payoff of disclosing at $t + \Delta_n$ is

$$V(t, \Delta_n) := \int_t^{t+\Delta_n} e^{-r_n(s-t)} p_{n,s} ds + e^{-r_n \Delta_n} \Pi(q_{n,t+\Delta_n}^G).$$

We make use of the following bounds. First, by the support restriction, for each n , $q_{n,t}^G \geq \phi_{t-t_0}(1) \geq e^{-\lambda_1(t-t_0)}$, since good evidence disclosed at time t has age at most $t - t_0$. Second, $q_{n,t+\Delta_n}^G \leq 1$. Third, by Lemma 22, there is a uniform bound (across equilibria, n , and time)

$p_{n,s} < p(\delta) < 1$. Incorporating these bounds yields

$$\begin{aligned}\Pi(q_{n,t}^G) &\geq \frac{p^*}{r_n} + \frac{e^{-\lambda_1(t-t_0)} - p^*}{\Lambda + r_n} \\ V(t, \Delta_n) &\leq \int_t^{t+\Delta_n} e^{-r_n(s-t)} p(\delta) ds + e^{-r_n\Delta_n} \Pi(1).\end{aligned}$$

Subtracting and scaling by r_n ,

$$r_n(V(t, \Delta_n) - \Pi(q_{n,t}^G)) \leq r_n \left(\int_t^{t+\Delta_n} e^{-r_n(s-t)} p(\delta) ds + e^{-r_n\Delta_n} \Pi(1) - \frac{p^*}{r_n} - \frac{e^{-\lambda_1(t-t_0)} - p^*}{\Lambda + r_n} \right).$$

Since $\Delta_n \geq \Delta_0 > 0$ for all n , the right hand side has limit $p(\delta) - e^{-\lambda_1(t-t_0)}$. By choosing t sufficiently close to t_0 at the outset, this bound is negative, and therefore for sufficiently large n , the strategy cannot be part of any equilibrium.

For the second part of the proposition, the time- t payoff from disclosing bad evidence is

$$\int_0^\infty e^{-rs} [p^* - (p^* - q_t^B) e^{-(\lambda_0 + \lambda_1)s}] ds = \frac{p^*}{r} - \frac{p^* - q_t^B}{r + \lambda_0 + \lambda_1}. \quad (44)$$

The payoff of disclosing at time $t + \epsilon$ for any $\epsilon > 0$ is at least

$$\int_0^\epsilon e^{-rs} p_{t+s} ds, \quad (45)$$

since the continuation flow utility after $t + \epsilon$ is nonnegative.

After subtracting (44) from (45) and scaling by r , the benefit of delay by ϵ is at least

$$\int_0^\epsilon r e^{-rs} p_{t+s} ds - p^* + \frac{r}{r + \lambda_0 + \lambda_1} (p^* - q_t^B). \quad (46)$$

By Lemma 18, this is at least

$$\int_0^\epsilon r e^{-rs} [p_t - (\lambda_1 + \mu)s] ds - p^* + \frac{r}{r + \lambda_0 + \lambda_1} (p^* - q_t^B).$$

As $r \rightarrow \infty$, this converges uniformly over t and bad evidence disclosure strategies to

$$p_t - p^* + (p^* - q_t^B) = p_t - q_t^B.$$

All that remains is to show that at all on-path bad evidence disclosure times $t \in [0, T]$, we have $p_t - q_t^B > 0$.

In an equilibrium with age-independent strategies, if disclosure is on path at time t , then

$q_t^B = \bar{q}_{t-}^B = \frac{B_{1,t-}^{un}}{B_t^{un}}$ at all times, with $q_0^B = 0$. We also have $p_t \geq p_{t-}$ by Lemma 18, where

$$p_{t-} = \frac{N_t^{un}\phi_t + B_{1,t-}^{un}}{N_t^{un} + B_{t-}^{un}}. \quad (47)$$

Hence,

$$p_t - q_t^B = p_t - \bar{q}_{t-}^B \geq p_{t-} - \bar{q}_{t-}^B = \frac{N_t^{un}}{N_t^{un} + B_{t-}^{un}}(\phi_t - \bar{q}_{t-}^B) \geq N_t^{un}(\phi_t - \bar{q}_{t-}^B) = e^{-\mu t}(\phi_t - \bar{q}_{t-}^B).$$

We also have $\bar{q}_{t-}^B \leq q_t^{B,0} = p^*(1 - e^{-\Lambda t})$ and $\phi_t = p^* - (p^* - p_0)e^{-\Lambda t}$, so

$$\phi_t - \bar{q}_{t-}^B \geq p_0 e^{-\Lambda t}.$$

Together, for all $t \in [0, T]$, these inequalities imply

$$p_t - q_t^B \geq e^{-\mu t} p_0 e^{-\Lambda t} \geq p_0 e^{-(\Lambda + \mu)T} > 0,$$

concluding the proof. □

Proof of Lemma 2. We prove the second part of the lemma first. More specifically, we state and prove the following lemma.

Lemma 19. *Assume $p_0 \in \left(\frac{\lambda_0}{\lambda_0 + \lambda_1 + r}, \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}\right)$ and that $\mu < \left(\frac{\lambda_0 + \lambda_1 + r}{\lambda_1 + r}\right)r$. Then there exists an equilibrium with immediate good evidence disclosure and no bad evidence disclosure.*

Proof of Lemma 19. We verify optimality of the sender's strategy assuming the receiver correctly conjectures immediate disclosure of good evidence and no disclosure of bad evidence, and that following off-path disclosure of bad evidence at time t , the receiver assumes the evidence arrived at time t .

Let $q_{B,t}$ (identically 0 under our off-path belief assumption) denote the receiver's posterior immediately after disclosure of bad evidence.

Some algebra shows that delaying bad evidence disclosure is locally incentive compatible if and only if

$$\dot{q}_{B,t} \geq (\lambda_0 + \lambda_1 + r)(q_{B,t} - p_t) + (\lambda_0 + \lambda_1)(p^* - q_{B,t}).$$

With $q_B \equiv 0$, this reduces to

$$p_t \geq \frac{\lambda_0}{\lambda_0 + \lambda_1 + r}, \quad (48)$$

and hence to establish that it is optimal to never disclose bad evidence, it suffices to show that p_t always satisfies this inequality. Note that by assumption, the inequality is satisfied strictly at time 0.

Let q_t^G (identically 1 under the conjectured strategy) denote the receiver's belief immediately after disclosure of good evidence. Delaying disclosure of good evidence is not locally incentive compatible if

$$\begin{aligned}\dot{q}_t^G &< (\lambda_0 + \lambda_1 + r)(q_t^G - p_t) + (\lambda_0 + \lambda_1)(p^* - q_t^G) \\ &= r q_t^G - (\lambda_0 + \lambda_1 + r)p_t + \lambda_0.\end{aligned}$$

With $q_G \equiv 1$, this reduces to

$$p_t < \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}. \quad (49)$$

Hence, whenever this inequality is satisfied, the sender prefers to disclose good evidence immediately rather than at any future time. Recall that this inequality is satisfied at time 0 by assumption.

We now show that under the condition on μ in the proposition, $p_t \in \left(\frac{\lambda_0}{\lambda_0 + \lambda_1 + r}, \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}\right)$ for all $t \geq 0$ conditional on no disclosure.

Let $B_{x,t}$ denote the probability that the sender has bad evidence and $\theta_t = x \in \{0, 1\}$, conditional on no disclosure yet. The laws of motion for (B_1, B_0, p) under the receiver's conjecture are

$$\dot{B}_{1,t} = B_{0,t}\lambda_0 - B_{1,t}\lambda_1 + B_{1,t}(p_t - B_{1,t})\mu. \quad (50)$$

$$\dot{B}_{0,t} = -B_{0,t}\lambda_0 + B_{1,t}\lambda_1 + (1 - p_t - B_{0,t})\mu + B_{0,t}(p_t - B_{1,t})\mu \quad (51)$$

$$\dot{p}_t = \lambda_0(1 - p_t) - \lambda_1 p_t - \mu(p_t - B_{1,t})(1 - p_t). \quad (52)$$

with $B_{0,0} = B_{1,0} = 0$.

Standard arguments using the comparison theorem show that there is a unique solution (B_1, B_0, p) to this system for all time, all variables lie in $(0, 1)$, and $p_t > B_{1,t}$. By inspection, the right hand side of (52) is negative when p_t exceeds $p^* = \frac{\lambda_0}{\lambda_0 + \lambda_1}$, which lies in $\left(\frac{\lambda_0}{\lambda_0 + \lambda_1 + r}, \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}\right)$. Hence, p_t cannot hit the upper end of this interval in finite time.

Next, note that the right hand side of (52) is bounded below by

$$f(p_t) := \lambda_0(1 - p_t) - \lambda_1 p_t - \mu(p_t - 0)(1 - p_t).$$

Let $\tilde{p}$ denote the solution to $\dot{\tilde{p}}_t = f(\tilde{p}_t)$ with initial condition $\tilde{p}_0 = p_0$. It is easy to show

that $p_t \geq \tilde{p}_t$ at all times, and that $\tilde{p}_t$ converges monotonically to a steady state $\tilde{p}^* \in (0, 1)$. The condition $\mu < \left(\frac{\lambda_0 + \lambda_1 + r}{\lambda_1 + r}\right) r$ ensures that this steady state is above $\frac{\lambda_0}{\lambda_0 + \lambda_1 + r}$. Therefore, p_t does not cross this threshold. $\square$

For the first part of Lemma 2, suppose the receiver conjectures immediate disclosure of good evidence and no disclosure of bad evidence. We state the explicit formulas for belief stock variables, which we refer to in later proofs as well:

$$B_{1,t} = \frac{\int_0^t \mu e^{-\mu s} (1 - \phi_s) q_t^{B,s} ds}{e^{-\mu t} + \int_0^t \mu e^{-\mu s} (1 - \phi_s) ds} \quad (53)$$

$$B_{0,t} = \frac{\int_0^t \mu e^{-\mu s} (1 - \phi_s) (1 - q_t^{B,s}) ds}{e^{-\mu t} + \int_0^t \mu e^{-\mu s} (1 - \phi_s) ds} \quad (54)$$

$$B_{1,t}^{un} = \int_0^t \mu e^{-\mu s} (1 - \phi_s) q_t^{B,s} ds \quad (55)$$

$$B_{0,t}^{un} = \int_0^t \mu e^{-\mu s} (1 - \phi_s) (1 - q_t^{B,s}) ds \quad (56)$$

$$B_t^{un} = \int_0^t \mu e^{-\mu s} (1 - \phi_s) ds \quad (57)$$

$$N_t^{un} = e^{-\mu t}. \quad (58)$$

Under neutral beliefs, the belief immediately after disclosure of bad evidence at time t is $q_t^B = \bar{q}_t^B = \frac{B_{1,t}}{B_{0,t} + B_{1,t}}$, where $B_{1,t}$ and $B_{0,t}$ are given by (53) and (54). We also have

$$p_t = \frac{\phi_t N_t^{un} + B_{1,t}^{un}}{N_t^{un} + B_t^{un}} \quad (59)$$

$$\bar{q}_t^B = \frac{B_{1,t}}{B_t} = \frac{B_{1,t}^{un}}{B_t^{un}}. \quad (60)$$

A useful identity is

$$p_t - \bar{q}_t^B = \frac{N_t^{un}}{N_t^{un} + B_t^{un}} (\phi_t - \bar{q}_t^B). \quad (61)$$

Note that this is positive, since $\bar{q}_t^B \leq q_t^0 < \phi_t$. Moreover, (B_1^{un}, B^{un}) satisfy the system

$$\dot{B}_{1,t}^{un} = \lambda_0 (B_t^{un} - B_{1,t}^{un}) - \lambda_1 B_{1,t}^{un} \quad (62)$$

$$\dot{B}_t^{un} = \mu e^{-\mu t} (1 - \phi_t). \quad (63)$$

Thus $\bar{q}^B$ satisfies

$$\dot{\bar{q}}_t^B = \frac{\dot{B}_{1,t}^{un}}{B_t^{un}} - \bar{q}_t^B \frac{\dot{B}_t^{un}}{B_t^{un}} = \lambda_0(1 - \bar{q}_t^B) - \lambda_1 \bar{q}_t^B - \bar{q}_t^B \frac{\dot{B}_t^{un}}{B_t^{un}}. \quad (64)$$

The local IC constraint for delaying disclosure is

$$\dot{q}_t^B \geq (\lambda_0 + \lambda_1 + r)(q_t^B - p_t) + (\lambda_0 + \lambda_1)(p^* - q_t^B) = r q_t^B - (\Lambda + r)p_t + \lambda_0. \quad (65)$$

Note that if this condition fails for all $s \geq t$, then disclosure at t is optimal. We now show that there exists such t .

Under neutral off-path beliefs, we have $q_t^B = \bar{q}_t^B$ at all times. Using this and substituting (64), (65) becomes

$$\begin{aligned} \lambda_0(1 - \bar{q}_t^B) - \lambda_1 \bar{q}_t^B - \bar{q}_t^B \frac{\dot{B}_t^{un}}{B_t^{un}} &\geq r \bar{q}_t^B - (\Lambda + r)p_t + \lambda_0 \\ \iff \bar{q}_t^B \frac{\dot{B}_t^{un}}{B_t^{un}} &\leq (\Lambda + r)(p_t - \bar{q}_t^B). \end{aligned} \quad (66)$$

By (61) and rearranging, this is equivalent to

$$\Lambda + r \geq \bar{q}_t^B \frac{\dot{B}_t^{un}}{B_t^{un}} \frac{N_t^{un} + B_t^{un}}{N_t^{un}(\phi_t - \bar{q}_t^B)} = \bar{q}_t^B \frac{\mu(1 - \phi_t)(N_t^{un} + B_t^{un})}{B_t^{un}(\phi_t - \bar{q}_t^B)}. \quad (67)$$

On the right hand side, assuming $\lambda_1, \lambda_0 > 0$, we have the following limits. First, $\phi_t \rightarrow p^* \in (0, 1)$, $N_t^{un} \rightarrow 0$, $B_t^{un} \rightarrow B_\infty^{un} := \int_0^\infty \mu e^{-\mu s} (1 - \phi_s) ds \in (0, 1)$, and for each s , $\lim_{t \rightarrow \infty} q_t^{B,s} = p^*$. Next, by dominated convergence, $B_{1,t}^{un} = \int_0^\infty \mathbb{1}_{s \leq t} \mu e^{-\mu s} (1 - \phi_s) q_t^{B,s} ds \rightarrow p^* \int_0^\infty \mu e^{-\mu s} (1 - \phi_s) ds = p^* B_\infty^{un}$. Hence, $\bar{q}_t^B \rightarrow p^*$, which implies $\phi_t - \bar{q}_t^B \rightarrow 0$ from above. Hence, (67) fails for sufficiently large t , as desired. $\square$

Proof of Theorem 3. We show that such an equilibrium exists for large r , supported by neutral off-path beliefs after bad evidence disclosure strictly between bursts.

We first establish optimality of immediately disclosing good evidence. Immediately after a burst, the receiver believes that the sender does not possess evidence, and thus beliefs coincide with $\phi_t = p^* = p_0$ given the stationary prior. In between bursts, the possibility that the sender possesses undisclosed bad evidence means that $p_t < p^*$. Hence, at all times we have $p_t \leq p^* < \frac{\lambda_0 + r}{\Lambda + r}$, and by the argument in the proof of Lemma 19, disclosing good evidence immediately is indeed optimal when it is conjectured.

We now show that for large r , there exists Δ such that the bad evidence disclosure policy is optimal when it is conjectured.

Suppose the bursts are conjectured to be spaced Δ units apart for some $\Delta > 0$. Immediately after any burst, normalizing the current time to 0, the belief variables evolve exactly as in (53)-(58) and (59)-(60).

The first optimality condition is that delay must be optimal between bursts. Under the neutral off-path beliefs, disclosing at time $t \in (0, \Delta)$ or at the next burst, $t = \Delta$, results in a posterior of $\bar{q}_t^B$. Hence, the local delay IC condition is still (65) or equivalently (66). At time 0, we have $\dot{q}_0^B > 0$ while the RHS of (65) reduces to $-(\lambda_0 + \lambda_1 + r)p^* + (\lambda_0 + \lambda_1)p^* = -rp^* < 0$. Hence, by continuity, delay is strictly optimal initially. Moreover, we have already seen that (66) fails for large t ; define $\bar{\Delta}_r$ as the infimum of such t :

$$\bar{\Delta}_r := \inf\{t > 0 : (66) \text{ fails}\}.$$

By definition, delay is locally incentive compatible in between bursts at regular intervals of length Δ if and only if $\Delta \leq \bar{\Delta}_r$.

The second optimality condition is that disclosure must be optimal at each burst. Suppose the time is $n\Delta$ and the sender has bad evidence. By disclosing it immediately, the sender obtains payoff calculated earlier as

$$\Pi(q_\Delta^B) := \frac{p^*}{r} - \frac{p^* - q_\Delta^B}{r + \lambda_0 + \lambda_1},$$

where we have used that $q_{n\Delta}^B = q_\Delta^B$ by the cyclical property of beliefs.

By waiting until $(n+1)\Delta$ to disclose, the sender obtains

$$\int_{n\Delta}^{(n+1)\Delta} e^{-r(s-n\Delta)} p_s ds + e^{-r\Delta} \Pi(q_\Delta^B).$$

Disclosure at time Δ is therefore optimal iff

$$\Gamma(\Delta) := \Pi(q_\Delta^B)(1 - e^{-r\Delta}) - \int_0^\Delta e^{-rs} p_s ds \geq 0, \quad (68)$$

where we have used that beliefs are cyclical with period Δ . Note that if the sender is willing to forego disclosure at a burst, he is willing to forego disclosure at all bursts, so the condition is equivalent to checking whether the sender prefers to disclose at a burst versus never disclose.

The condition (68) fails for all sufficiently small Δ . In particular, we have equality at $\Delta = 0$ while the derivatives of the left and right hand sides with respect to Δ at $\Delta = 0$ are $r\Pi$ and p^* , respectively, where $\Pi < p^*/r$. However, (68) holds for all sufficiently large Δ ;

taking $\Delta \rightarrow \infty$, we have $q_\Delta^B \rightarrow p^*$ and therefore $\Pi(q_\Delta^B) \rightarrow \Pi(p^*) = p^*/r$, while the value of concealing forever is $\int_0^\infty e^{-rs} p_s ds < \int_0^\infty e^{-rs} p^* ds = p^*/r$.

The rest of the proof consists of showing that for sufficiently large r , $\Gamma(\bar{\Delta}_r) > 0$, so that our conjectured strategy profile with $\Delta = \bar{\Delta}_r$ is an equilibrium.

Let $y_t := p^* - p_t$. For later use, note that $y_0 = 0$, $\dot{y}_0 = \mu p^*(1 - p^*)$, and $\ddot{y}_0 = \mu p^*(1 - p^*)(\mu(2p^* - 1) - \Lambda)$. Then $\Gamma(\bar{\Delta}_r)$ rearranges to

$$\int_0^{\bar{\Delta}_r} e^{-rs} y_s ds - (1 - e^{-r\bar{\Delta}_r}) \frac{p^* - \bar{q}_{\bar{\Delta}_r}^B}{\Lambda + r}$$

Lemma 20. *As $r \rightarrow \infty$, we have $\bar{\Delta}_r \rightarrow +\infty$, $r(p^* - \bar{q}_{\bar{\Delta}_r}^B) \rightarrow \dot{y}_0$, and $r^2\Gamma(\bar{\Delta}_r) \rightarrow 0$.*

In light of Lemma 20, to show eventual positivity of $\Gamma(\bar{\Delta}_r)$, we show that

$$\lim_{r \rightarrow \infty} r^3\Gamma(\bar{\Delta}_r) > 0.$$

Now $r^3\Gamma(\bar{\Delta}_r)$ expands as

$$r^3\Gamma(\bar{\Delta}_r) = r^3 \int_0^{\bar{\Delta}_r} e^{-rs} y_s ds - r^3(1 - e^{-r\bar{\Delta}_r}) \frac{p^* - \bar{q}_{\bar{\Delta}_r}^B}{\Lambda + r}. \quad (69)$$

Since the limit $r^2\Gamma_{\bar{\Delta}_r}$ had the form $\ddot{y}_0 - \dot{y}_0$, taking limits on each term alone would yield $\infty - \infty$; so instead, we recenter by $\dot{y}_0$:

$$r^3\Gamma_{\bar{\Delta}_r} = r^3 \underbrace{\left(\int_0^{\bar{\Delta}_r} e^{-rs} y_s ds - \frac{\dot{y}_0}{r^2} \right)}_{:=X(r)} - r^3 \underbrace{\left((1 - e^{-r\bar{\Delta}_r}) \frac{p^* - \bar{q}_{\bar{\Delta}_r}^B}{\Lambda + r} - \frac{\dot{y}_0}{r^2} \right)}_{:=Y(r)}. \quad (70)$$

Lemma 21. *As $r \rightarrow \infty$, $X(r) \rightarrow \ddot{y}_0 = \dot{y}_0(\mu(2p^* - 1) - \Lambda)$ and $Y(r) \rightarrow \dot{y}_0(\max\{\Lambda - \mu, 0\} - 2\Lambda - \mu(1 - p^*)) < \ddot{y}_0$. Hence, $\lim_{r \rightarrow \infty} r^3\Gamma_{\bar{\Delta}_r} > 0$.*

Given Lemma 21, $\Gamma(\bar{\Delta}_r) > 0$ for sufficiently large $r > 0$, so the construction is complete. $\square$

Proof of Lemma 3. Since bad evidence is disclosed immediately at all $t \geq t_0$, all bad evidence disclosed strictly after t_0 is assumed to be fresh: $q_t^B \equiv 0$ for all $t > t_0$. This implies that after disclosing bad evidence at any $t > t_0$, the sender's flow payoff at each $s \geq t$ is $\phi_{s-t}(0)$. Moreover, conditional on nondisclosure after t_0 , the receiver believes that with probability 1, the sender does not possess evidence of either type. Hence $p_t \equiv \phi_t$ for all $t > t_0$. It follows

that by never disclosing, the sender’s flow payoff at each $s \geq t$ is $\phi_s = \phi_{s-t}(\phi_t) > \phi_{s-t}(0)$. Thus, disclosing bad evidence at t is not optimal for any $t > t_0$. $\square$

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B Online Appendix

B.1 Supporting results for Section 4.3

Proof of Lemma 18. Let π_N , π_G , and π_B denote the probability of no disclosure, good evidence disclosure, and bad evidence disclosure, respectively, on $(t, t + s]$ conditional on no disclosure on $[0, t]$. Let $\phi_s(p_t)$ denote the belief at time $t + s$ based on the belief p_t at time t and no information other than Markov switching dynamics in $(t, t + s]$. By the law of iterated expectations,

$$\phi_s(p_t) = \pi_N p_{t+s} + \pi_B \mathbb{E}[\theta_{t+s} | \sigma^B \in (t, t + s]] + \pi_G \mathbb{E}[\theta_{t+s} | \sigma^G \in (t, t + s]].$$

We claim that $\mathbb{E}[\theta_{t+s} | \sigma^B \in (t, t + s]] \leq \phi_s(\bar{q}_t^B) \leq \phi_s(p_t)$. To understand the first inequality, write

$$\begin{aligned} \mathbb{E}[\theta_{t+s} | \sigma^B \in (t, t + s]] &= \Pr(\tau \leq t | \sigma^B \in (t, t + s]) \mathbb{E}[\theta_{t+s} | \tau \leq t, \sigma^B \in (t, t + s]] \\ &\quad + \Pr(\tau > t | \sigma^B \in (t, t + s]) \mathbb{E}[\theta_{t+s} | \tau > t, \sigma^B \in (t, t + s]]. \end{aligned}$$

For the first term, since bad evidence disclosure is age-independent, we have $\mathbb{E}[\theta_{t+s} | \tau \leq t, \sigma^B \in (t, t + s]] = \phi_s(\bar{q}_t^B)$. For the second term, if evidence arrives at some time $u \in (t, t + s]$, then $\theta_u = 0$, and the conditional belief about θ_{t+s} is $\phi_{t+s-u}(0) \leq \phi_s(0) \leq \phi_s(\bar{q}_t^B)$. Hence, averaging over u , we have $\mathbb{E}[\theta_{t+s} | \tau > t, \sigma^B \in (t, t + s]] \leq \phi_s(\bar{q}_t^B)$ and overall $\mathbb{E}[\theta_{t+s} | \sigma^B \in (t, t + s]] \leq \phi_s(\bar{q}_t^B) \leq \phi_s(p_t)$, where the last inequality is due to $\bar{q}_t^B \leq p_t$.

Also, $\mathbb{E}[\theta_{t+s} | \sigma^G \in (t, t + s]] \leq 1$. Rearranging the equation and using these inequalities, we have

$$\begin{aligned} \pi_N p_{t+s} &\geq \phi_s(p_t) - \pi_B \phi_s(p_t) - \pi_G \\ \iff \pi_N (p_{t+s} - p_t) &\geq (\pi_N + \pi_G)(\phi_s(p_t) - p_t) - \pi_G(1 - p_t). \end{aligned}$$

Now $\pi_N > 0$ and we have

$$\begin{aligned} p_{t+s} - p_t &\geq \left(1 + \frac{\pi_G}{\pi_N}\right) (\phi_s(p_t) - p_t) - \frac{\pi_G}{\pi_N} (1 - p_t) \\ &= (\phi_s(p_t) - p_t) - \frac{\pi_G}{\pi_N} (1 - \phi_s(p_t)). \end{aligned} \tag{71}$$

Expanding the first term yields

$$\begin{aligned}\phi_s(p_t) - p_t &= p^* - (p^* - p_t)e^{-(\lambda_0 + \lambda_1)s} - p_t = (p^* - p_t)(1 - e^{-(\lambda_0 + \lambda_1)s}) \\ &\geq -(1 - p^*)(1 - e^{-(\lambda_0 + \lambda_1)s}) = -\frac{\lambda_1}{\lambda_0 + \lambda_1}(1 - e^{-(\lambda_0 + \lambda_1)s}) \geq -\lambda_1 s.\end{aligned}$$

Let M be the probability that no evidence has arrived in $[0, t]$ conditional on no disclosure yet in $[0, t]$. We have $\pi_G \leq M(1 - e^{-\mu s})$ and $\pi_N \geq Me^{-\mu s}$, and therefore $\frac{\pi_G}{\pi_N} \leq \frac{1 - e^{-\mu s}}{e^{-\mu s}} = e^{\mu s} - 1$. Hence, returning to (71),

$$p_{t+s} - p_t \geq -\lambda_1 s - (e^{\mu s} - 1).$$

Since we can partition $[t, t + s]$ into N intervals of width s/N and apply the same bound on each interval, summing, and taking N arbitrarily large, we have

$$p_{t+s} - p_t \geq -(\lambda_1 + \mu)s.$$

To derive the implication $p_t \geq p_{t-}$, fix $u > 0$, set $t = u - s$, and take $s \rightarrow 0$. Then the previous result and existence of a left limit imply $p_u \geq p_{u-s} - (\lambda_1 + \mu)s \rightarrow p_{u-}$. $\square$

The following lemma states a useful upper bound on p_t . Intuitively, for p_t to be close to 1, it must be that the sender possesses very recent evidence with very high probability, which is impossible.

Lemma 22. *Fix any $\delta > 0$. Then in all equilibria, p_t is uniformly bounded away from 1: $p_t \leq p(\delta) := e^{-\mu\delta} \max\{p_0, p^* + (1 - p^*)e^{-\Lambda\delta}\} + (1 - e^{-\mu\delta}) \cdot 1 < 1$.*

Proof. Suppose there has been no disclosure up to time t . By definition, the belief is p_t . Now fix any $\delta > 0$ (allowing for $\delta > t$). There are four possibilities: (i) no evidence has arrived by t , (ii) good evidence arrived by $t - \delta$, (iii) good evidence arrived in $(t - \delta, t]$, and (iv) bad evidence arrived. In case (i), the updated belief is $\phi_t \leq \max\{p_0, p^*\}$. In case (ii), it is at most $\phi_\delta(1) = p^* + (1 - p^*)e^{-\Lambda\delta}$. In case (iii), it is at most 1. In case (iv), it is at most $\phi_t(0) = p^*(1 - e^{-\Lambda t}) \leq p^*$. Let A be the probability of (iii) conditional on no disclosure up to time t . Then we have

$$p_t \leq (1 - A) \max\{p_0, p^* + (1 - p^*)e^{-\Lambda\delta}\} + A \cdot 1. \quad (72)$$

Also, by Bayes' rule we have

$$\begin{aligned}
A &= \frac{\Pr(\tau^G \in (t - \delta, t], \sigma > t)}{\Pr(\sigma > t)} \\
&\leq \frac{\Pr(\tau \in (t - \delta, t], \sigma > t)}{\Pr(\sigma > t)} \\
&= \frac{\Pr(\tau \in (t - \delta, t], \sigma > t)}{\Pr(\tau \in (t - \delta, t], \sigma > t) + \Pr(\tau \leq t - \delta, \sigma > t) + \Pr(\tau > t)} \\
&\leq \frac{\Pr(\tau \in (t - \delta, t], \sigma > t)}{\Pr(\tau \in (t - \delta, t], \sigma > t) + \Pr(\tau > t)} \\
&\leq \frac{e^{-\mu(t-\delta)} - e^{-\mu t}}{e^{-\mu(t-\delta)} - e^{-\mu t} + e^{-\mu t}} \\
&= 1 - e^{-\mu\delta},
\end{aligned}$$

where the second to last line uses that $\Pr(\tau \in (t - \delta, t]) = e^{-\mu(\max\{t-\delta, 0\}} - e^{-\mu t} \leq e^{-\mu(t-\delta)} - e^{-\mu t}$ for all $\delta > 0$, including $\delta > t$. Using this in the inequality for p_t yields the uniform upper bound in the lemma. $\square$

Proof of Lemma 20. The fact that $\bar{\Delta}_r \rightarrow \infty$ can be seen in (66), where r only enters the term $(\Lambda + r)$, and so on any finite interval, the inequality holds for all sufficiently large r .

Next, we show that $\lim_{r \rightarrow \infty} r^2 \Gamma_{\bar{\Delta}_r} = 0$. We have

$$r^2 \Gamma_{\bar{\Delta}_r} = r^2 \int_0^\epsilon e^{-rs} y_s ds + r^2 \int_\epsilon^{\bar{\Delta}_r} e^{-rs} y_s ds - r^2 (1 - e^{-r\bar{\Delta}_r}) \frac{p^* - \bar{q}_{\bar{\Delta}_r}^B}{\Lambda + r}. \quad (73)$$

On the right hand side of (73), the second term clearly tends to 0 as $r \rightarrow \infty$. The first term Taylor-expands as

$$\begin{aligned}
&r^2 \int_0^\epsilon e^{-rs} [y_0 + s\dot{y}_0 + \frac{s^2}{2}\ddot{y}_0 + o(s^2)] ds \\
&= \dot{y}_0 [1 - e^{-r\epsilon}(1 + r\epsilon)] + \frac{\ddot{y}_0}{2r} [2 - e^{-r\epsilon}(2 + r\epsilon(2 + r\epsilon))] + r^2 \int_0^\epsilon e^{-rs} o(s^2) ds \\
&\rightarrow \dot{y}_0 + 0 + 0 = \dot{y}_0.
\end{aligned}$$

The last term in (73) has the same limit as $r(p^* - \bar{q}_{\bar{\Delta}_r}^B)$ as $r \rightarrow \infty$ because $\frac{r}{\Lambda + r}(1 - e^{-r\bar{\Delta}_r}) \rightarrow 1$ (due to $\bar{\Delta}_r \rightarrow \infty$). Because of the stationary prior, $p^* = \phi_{\bar{\Delta}_r}$, and then since local IC (67) binds at $\bar{\Delta}_r$, we have

$$r(p^* - \bar{q}_{\bar{\Delta}_r}^B) = \frac{r}{\Lambda + r} \bar{q}_{\bar{\Delta}_r}^B \frac{\mu(1 - p^*)(N_{\bar{\Delta}_r}^{un} + B_{\bar{\Delta}_r}^{un})}{B_{\bar{\Delta}_r}^{un}} \rightarrow \mu p^*(1 - p^*) = \dot{y}_0, \quad (74)$$

where we have used that $N_{\bar{\Delta}_r}^{un} = e^{-\mu\bar{\Delta}_r}$, that $\bar{\Delta}_r \rightarrow \infty$, and the limits derived in the proof of Lemma 2. Thus, the last term in (73) tends to $\dot{y}_0$, so $\lim_{r \rightarrow \infty} r^2 \Gamma_{\bar{\Delta}_r} = \dot{y}_0 + 0 - \dot{y}_0 = 0$. $\square$

Proof of Lemma 21. For $X(r)$, we split the integral into $[0, \epsilon)$ and $[\epsilon, \Delta)$ and repeat the Taylor expansion from the proof of Lemma 20:

$$\int_0^{\bar{\Delta}_r} e^{-rs} y_s ds = \int_0^\epsilon e^{-rs} [y_0 + \dot{y}_0 s + \frac{1}{2} \ddot{y}_0 s^2 + o(s^2)] ds + \int_\epsilon^{\bar{\Delta}_r} e^{-rs} y_s ds. \quad (75)$$

We have $r^3 \int_\epsilon^{\bar{\Delta}_r} e^{-rs} y_s ds \rightarrow 0$ as $r \rightarrow \infty$, so $X(r)$ has the same limit as

$$r^3 \left(\int_0^\epsilon e^{-rs} [y_0 + \dot{y}_0 s + \frac{1}{2} \ddot{y}_0 s^2 + o(s^2)] ds - \frac{\dot{y}_0}{r^2} \right). \quad (76)$$

Recall that $y_0 = 0$ and $\dot{y}_0 = -\dot{p}_0$. Hence (76) simplifies to

$$r^3 \dot{y}_0 \left(\int_0^\epsilon e^{-rs} s ds - \frac{1}{r^2} \right) + r^3 \int_0^\epsilon e^{-rs} \frac{1}{2} \ddot{y}_0 s^2 ds + r^3 \int_0^\epsilon e^{-rs} o(s^2) ds. \quad (77)$$

We address each term in (77) individually. The first term reduces to

$$r^3 \dot{y}_0 \left(\frac{1}{r^2} (1 - (1 + r\epsilon)e^{-r\epsilon}) - \frac{1}{r^2} \right) = -\dot{y}_0 r (1 + r\epsilon) e^{-r\epsilon} \xrightarrow{r \rightarrow \infty} 0. \quad (78)$$

The second term in (77) gives

$$\frac{1}{2} \ddot{y}_0 [2 - (2 + 2r\epsilon + r^2 \epsilon^2) e^{-r\epsilon}] \rightarrow \ddot{y}_0. \quad (79)$$

Finally, the third limit can be made arbitrarily small by taking ϵ sufficiently small. We conclude that $\lim_{r \rightarrow \infty} X(r) = \ddot{y}_0 = \dot{y}_0(\mu(2p^* - 1) - \Lambda)$.

For $Y(r)$, rearrange it to

$$r^3 (1 - e^{-r\bar{\Delta}_r}) \left(\frac{p^* - \bar{q}_{\bar{\Delta}_r}^B}{\Lambda + r} - \frac{\dot{y}_0}{r^2} \right) - r^3 e^{-r\bar{\Delta}_r} \frac{\dot{y}_0}{r^2}.$$

The final term tends to 0 and $1 - e^{-r\bar{\Delta}_r} \rightarrow 1$, leaving us to calculate the limit of

$$\begin{aligned}
r^3 \left(\frac{p^* - \bar{q}_{\Delta_r}^B}{\Lambda + r} - \frac{\dot{y}_0}{r^2} \right) &= \dot{y}_0 r^3 \left(\frac{\bar{q}_{\Delta_r}^B (N_{\Delta_r}^{un} + B_{\Delta_r}^{un})}{(\Lambda + r)^2 B_{\Delta_r}^{un} p^*} - \frac{1}{r^2} \right) \\
&= \dot{y}_0 r \frac{r^2 \bar{q}_{\Delta_r}^B (N_{\Delta_r}^{un} + B_{\Delta_r}^{un}) - (\Lambda + r)^2 B_{\Delta_r}^{un} p^*}{(\Lambda + r)^2 B_{\Delta_r}^{un} p^*} \\
&= \dot{y}_0 \left[\frac{r^3 (\bar{q}_{\Delta_r}^B (N_{\Delta_r}^{un} + B_{\Delta_r}^{un}) - B_{\Delta_r}^{un} p^*)}{(\Lambda + r)^2 B_{\Delta_r}^{un} p^*} - \frac{2\Lambda r^2}{(\Lambda + r)^2} - \frac{r\Lambda^2}{(\Lambda + r)^2} \right] \quad (80)
\end{aligned}$$

where the first equality uses the expression for $r(p^* - \bar{q}_{\Delta_r}^B)$ in (74) and that $\mu(1 - p^*) = \frac{\dot{y}_0}{p^*}$, the second simply combines fractions and simplifies, and the third splits by degree of r . Now the second and third terms in the square brackets in (80) contribute $-2\Lambda\dot{y}_0 - 0$ to the limit as $r \rightarrow \infty$. The first term in (80) contributes

$$\dot{y}_0 \frac{r^3 B_{\Delta_r}^{un} (\bar{q}_{\Delta_r}^B - p^*)}{(\Lambda + r)^2 B_{\Delta_r}^{un} p^*} + \dot{y}_0 \frac{r^3 \bar{q}_{\Delta_r}^B N_{\Delta_r}^{un}}{(\Lambda + r)^2 B_{\Delta_r}^{un} p^*}. \quad (81)$$

Since $r(\bar{q}_{\Delta_r}^B - p^*) \rightarrow -\dot{y}_0$ by Lemma 20, the first term of (81) tends to $-\frac{(\dot{y}_0)^2}{p^*}$. The second term of (81) has the same limit as $\dot{y}_0 r \frac{N_{\Delta_r}^{un}}{B_{\Delta_r}^{un}}$, because $\bar{q}_{\Delta_r}^B \rightarrow p^*$. We now calculate the limit of

$$r \frac{N_{\Delta_r}^{un}}{B_{\Delta_r}^{un}} = r(p^* - \bar{q}_{\Delta_r}^B) \frac{\frac{N_{\Delta_r}^{un}}{B_{\Delta_r}^{un}}}{p^* - \bar{q}_{\Delta_r}^B},$$

where again $r(p^* - \bar{q}_{\Delta_r}^B) \rightarrow \dot{y}_0$. Since r only enters the remaining term via $\bar{\Delta}_r$, we are only left with calculating

$$L := \lim_{t \rightarrow \infty} \frac{N_t^{un}}{B_t^{un}(p^* - \bar{q}_t^B)}, \quad (82)$$

for which we turn to closed forms. For $\Lambda \neq \mu$, under the stationary prior, we have

$$q_t^B = p^* \left(1 + \frac{\mu}{\Lambda - \mu} \frac{e^{-\Lambda t} - e^{-\mu t}}{1 - e^{-\mu t}} \right) \quad \text{and} \quad \frac{N_t^{un}}{B_t^{un}} = \frac{e^{-\mu t}}{(1 - p^*)(1 - e^{-\mu t})}.$$

Thus

$$\frac{N_t^{un}}{B_t^{un}(p^* - \bar{q}_t^B)} = \frac{e^{-\mu t}}{-(1 - p^*)(1 - e^{-\mu t}) p^* \frac{\mu}{\Lambda - \mu} \frac{e^{-\Lambda t} - e^{-\mu t}}{1 - e^{-\mu t}}} = -\frac{\Lambda - \mu}{\dot{y}_0(e^{-(\Lambda - \mu)t} - 1)}. \quad (83)$$

If $\Lambda > \mu$, we have $L = \frac{\Lambda - \mu}{\dot{y}_0}$. If $\Lambda < \mu$, $L = 0$. For the knife-edge case $\Lambda = \mu$, we have $\frac{N_t^{un}}{B_t^{un}(p^* - \dot{q}_t^B)} = \frac{1}{\dot{y}_0 t}$, so again $L = 0$. Hence, in general, $L = \frac{\max\{\Lambda - \mu, 0\}}{\dot{y}_0}$.

Retracing our steps, we have $\lim_{r \rightarrow \infty} Y(r) = -\frac{(\dot{y}_0)^2}{p^*} + (\dot{y}_0)^2 \frac{\max\{\Lambda - \mu, 0\}}{\dot{y}_0} - 2\Lambda \dot{y}_0$, where

$$\begin{aligned} \lim_{r \rightarrow \infty} (X(r) - Y(r)) &= \dot{y}_0(\mu(2p^* - 1) - \Lambda) - \left[-\frac{(\dot{y}_0)^2}{p^*} + \dot{y}_0 \max\{\Lambda - \mu, 0\} - 2\Lambda \dot{y}_0 \right] \\ &= \dot{y}_0 [\mu p^* + \Lambda - \max\{\Lambda - \mu, 0\}] > 0. \end{aligned}$$

□

B.2 Proof of Proposition 4

The proof has two main steps. First, we begin by proving the following corollary to Lemma 19, which establishes optimality of non-disclosure of bad evidence; it is stronger than needed for our purposes because it allows for arbitrary good evidence disclosure strategies, whereas our construction will involve a particular (three-phase) good evidence disclosure strategy. Second, we use a continuity argument to extend the good evidence disclosure strategy of Theorem 2.

For the first step, we use the following intermediate result.

Lemma 23. *Fix an arbitrary strategy H such that bad evidence is never disclosed. Let H^0 denote the strategy under which good evidence is disclosed immediately and bad evidence is never disclosed. Let p and p^0 denote the no-disclosure belief paths under H and H^0 , respectively. We claim that for all $t \geq 0$, $p_t \geq p_t^0$.*

Proof of Lemma 23. The proof uses similar logic to that of Proposition 2. Fix $t \geq 0$. For strategy H , define N as the ex ante probability that no evidence is disclosed by time t . For each $s \in [0, t]$, let ν_s^G denote the probability that good evidence arriving at time s is disclosed by time t . Let $\nu_s^{G,0} \equiv 1$ and N^0 have analogous definitions to ν_s^G and N , but under strategy H^0 instead of H .

We now apply the law of iterated expectations to H and H^0 . Even though timestamps are not observed by the receiver, we can condition on the true evidence-arrival time in the law-of-iterated-expectations decomposition. Thus,

$$\phi_t = N p_t + \int_0^t \mu \phi_s e^{-\mu s} \nu_s^G q_t^{G,s} ds \quad \text{and} \quad \phi_t = N^0 p_t^0 + \int_0^t \mu \phi_s e^{-\mu s} \nu_s^{G,0} q_t^{G,s} ds,$$

where we have used that bad evidence is not disclosed under either H or H^0 .

Equating the right hand sides and subtracting p_t^0 throughout yields

$$\begin{aligned} N(p_t - p_t^0) + \int_0^t \mu \phi_s e^{-\mu s} \nu_s^G (q_t^{G,s} - p_t^0) ds &= 0 + \int_0^t \mu \phi_s e^{-\mu s} \nu_s^{G,0} (q_t^{G,s} - p_t^0) ds \\ \implies N(p_t - p_t^0) &= \int_0^t \mu \phi_s e^{-\mu s} (\nu_s^{G,0} - \nu_s^G) (q_t^{G,s} - p_t^0) ds. \end{aligned}$$

Since $N \geq e^{-\mu t} > 0$ and $1 = \nu_s^{G,0} \geq \nu_s^G$ for all s and $q_t^{G,s} \geq \phi_t \geq p_t^0$, we conclude that $p_t \geq p_t^0$. $\square$

Corollary 3. *For sufficiently small $\lambda_0 > 0$, if the receiver conjectures that bad evidence is never disclosed and after off-path disclosure of bad evidence the receiver updates to 0, then it is optimal for the sender to never disclose bad evidence, independent of the good-evidence disclosure strategy.*

Proof of Corollary 3. From the proof of Lemma 19, under the given off-path beliefs, delaying disclosure of bad evidence is locally incentive compatible if $p_t \geq a(\lambda_0) := \frac{\lambda_0}{\lambda_0 + \lambda_1 + r}$ for all $t \geq 0$. Hence, it suffices to show that if $\lambda_0 > 0$ is sufficiently small, this inequality holds at all times for all possible good-evidence disclosure strategies. Given Lemma 23, we only need to show that $p_t^0 \geq a(\lambda_0)$ for all $t \geq 0$.

First, fix $\epsilon_0 > 0$. We show that there exists $T_0 > 0$ such that for all $\lambda_0 \in (0, \epsilon_0)$ and $t \geq T_0$, $\bar{q}_t^B$ cannot cross $a(\lambda_0)$ from above, where $\bar{q}_t^B$ is calculated under the strategy of immediate good disclosure and no bad disclosure. From (64), $\dot{\bar{q}}_t^B|_{\bar{q}_t^B = a(\lambda_0)} = \lambda_0(1 - \bar{q}_t^B) - \lambda_1 \bar{q}_t^B - \bar{q}_t^B \frac{\dot{B}_t^{un}}{B_t^{un}} = a(\lambda_0) \left(r - \frac{\dot{B}_t^{un}}{B_t^{un}} \right)$. Now define $\bar{\phi} := \max\{p_0, \frac{\epsilon_0}{\epsilon_0 + \lambda_1}\} < 1$. Then for all $\lambda_0 \in (0, \epsilon_0)$ and all t , we have $\phi_t \leq \max\{p_0, \frac{\lambda_0}{\lambda_0 + \lambda_1}\} \leq \bar{\phi}$. Thus, from (57),

$$\frac{\dot{B}_t^{un}}{B_t^{un}} = \frac{\mu e^{-\mu t} (1 - \phi_t)}{\int_0^t \mu e^{-\mu s} (1 - \phi_s) ds} \leq \frac{\mu e^{-\mu t}}{(1 - \bar{\phi})(1 - e^{-\mu t})}. \quad (84)$$

The right hand side of (84) is independent of λ_0 and tends to 0 as $t \rightarrow \infty$. We conclude that for all $t > T_0$, $\bar{q}_t^B$ cannot cross $a(\lambda_0)$ from above.

Next, we show that if $T \geq T_0$ is sufficiently large and $\epsilon_1 \leq \epsilon_0$ sufficiently small, $\bar{q}_T^B > a(\lambda_0)$

for all $\lambda_0 \in (0, \epsilon_1)$. This follows from

$$\begin{aligned} \frac{\bar{q}_T^B}{a(\lambda_0)} &= \frac{\frac{\lambda_0}{\Lambda} \int_0^T \mu e^{-\mu s} (1 - \phi_s) (1 - e^{-\Lambda(T-s)}) ds}{\frac{\lambda_0}{\Lambda+r} \int_0^T \mu e^{-\mu s} (1 - \phi_s) ds} \\ &\xrightarrow{\lambda_0 \rightarrow 0} \frac{\lambda_1 + r}{\lambda_1} \frac{\int_0^T \mu e^{-\mu s} (1 - p_0 e^{-\lambda_1 s}) (1 - e^{-\lambda_1(T-s)}) ds}{\int_0^T \mu e^{-\mu s} (1 - p_0 e^{-\lambda_1 s}) ds} \\ &\xrightarrow{T \rightarrow \infty} \frac{\lambda_1 + r}{\lambda_1} > 1. \end{aligned}$$

The claim then follows by choosing $T \geq T_0$ and then choosing ϵ_1 accordingly. Since we have ruled out crossing from above, and also $p_t^0 \geq \bar{q}_t^B$ by (61), this establishes that $p_t^0 \geq \bar{q}_t^B \geq a(\lambda_0)$ for all $t \geq T$ and $\lambda_0 \in (0, \epsilon_1)$.

Finally, we prove there exists $\epsilon_2 \leq \epsilon_1$ such that for all $\lambda_0 \in (0, \epsilon_2)$, $p_t^0 > a(\lambda_0)$ on $[0, T]$. Recall from the proof of Lemma 19 that p_t^0 is bounded below by the solution $\tilde{p}$ to

$$\dot{p}_t = \lambda_0(1 - p_t) - \lambda_1 p_t - \mu p_t(1 - p_t), \quad (85)$$

which by the comparison theorem is bounded below by the solution $p_t^b = p_0 e^{-(\lambda_1 + \mu)t}$ to

$$\dot{p}_t = -\lambda_1 p_t - \mu p_t. \quad (86)$$

For sufficiently small $\epsilon_2 > 0$, we have $p_T^b > a(\epsilon_2)$. It follows that for all $\lambda_0 \in (0, \epsilon_2)$ and all $t \in [0, T]$,

$$p_t \geq p_t^0 \geq p_t^b \geq p_T^b > a(\epsilon_2) > a(\lambda_0). \quad (87)$$

We have therefore established that $p_t \geq a(\lambda_0)$ for all $t \geq 0$ and all $\lambda_0 \in (0, \epsilon_2)$. □

For the second step, we now characterize the disclosure of good evidence and prove its optimality. During the stockpiling and purging phases, the laws of motion are

$$\dot{G}_{1,t} = G_{0,t} \lambda_0 - G_{1,t} \lambda_1 + (p_t - G_{1,t} - B_{1,t}) \mu - G_{1,t} \kappa_t (1 - G_{1,t} - G_{0,t}) \quad (88)$$

$$\dot{G}_{0,t} = -G_{0,t} \lambda_0 + G_{1,t} \lambda_1 - G_{0,t} \kappa_t (1 - G_{1,t} - G_{0,t}) \quad (89)$$

$$\dot{B}_{1,t} = B_{0,t} \lambda_0 - B_{1,t} \lambda_1 + B_{1,t} \kappa_t (G_{0,t} + G_{1,t}) \quad (90)$$

$$\dot{B}_{0,t} = -B_{0,t} \lambda_0 + B_{1,t} \lambda_1 + (1 - p_t - G_{0,t} - B_{0,t}) \mu + B_{0,t} \kappa_t (G_{0,t} + G_{1,t}). \quad (91)$$

$$\dot{p}_t = \lambda_0(1 - p_t) - \lambda_1 p_t - G_{1,t} \kappa_t + p_t (G_{0,t} + G_{1,t}) \kappa_t. \quad (92)$$

During the stockpiling phase, the local delay IC condition becomes

$$\dot{q}_t^G \geq (\lambda_0 + \lambda_1 + r)(q_t^G - p_t) + (\lambda_0 + \lambda_1)(p^* - q_t^G) \quad (93)$$

$$= r q_t^G - (\lambda_0 + \lambda_1 + r)p_t + \lambda_0. \quad (94)$$

Define $c(t; \lambda_0) = \dot{q}_t^G - [r q_t^G - (\lambda_0 + \lambda_1 + r)p_t + \lambda_0]$, generalizing $c(t)$ from the $\lambda_0 = 0$ case.

By repeating the arguments from the proof of Lemma 13, allowing for $\lambda_0 > 0$, we obtain cutoffs

$$\begin{aligned} \bar{p} &:= \frac{\lambda_0 + r + \frac{\lambda_1}{2}}{\lambda_0 + \lambda_1 + r} \\ \underline{p} &:= \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}. \end{aligned}$$

Specifically, within the class of three-phase equilibria with no disclosure of bad evidence, Lemma 13 goes through with these updated formulas for $\bar{p}$ and $\underline{p}$.

For $p_0 > \bar{p}$, we extend the definition of $\underline{T}$ as the first time t at which $c(t; \lambda_0) = 0$, and $\underline{T} = 0$ otherwise. Since the equations (88)-(92) and c are continuous in λ_0 and as we already showed $\frac{\partial}{\partial t} c(\underline{T}; 0) < 0$, $\underline{T}$ is continuous, and so are $(p_{\underline{T}}, q_{\underline{T}}^G, B_{1,\underline{T}}, B_{0,\underline{T}})$. After solving for $S(p, q^G, B_1)$ and $\kappa(p, q^G, B_1, B_0)$ in the same way as in the proof of Theorem 2, we obtain a system for (p, q^G, B_1, B_0) in the purging phase with initial conditions $(p_{\underline{T}}, q_{\underline{T}}^G, B_{1,\underline{T}}, B_{0,\underline{T}})$. This system and initial conditions are also continuous in λ_0 and are locally Lipschitz in (p, q^G, B_1, B_0) , so the unique solution is continuous in λ_0 near 0. The feasible set is then defined by the inequalities $0 \leq B_{1,t} < p_t < q_t^G < 1$ which ensure that S and κ are well-defined, and $A(p, q^G, B_1, B_0; \lambda_0) > 0$, where A is continuous and coincides with A from the proof of Theorem 2 at $\lambda_0 = 0$, where it was bounded away from 0. A similar first crossing argument using the comparison theorem establishes that the solution can only exit through $q_t^G = 1$. Moreover, we still have that $\delta_t = q_t^G - p_t$ satisfies (9), implying that this exit occurs at some finite time $\bar{T}$. After $\bar{T}$, immediate disclosure is incentive compatible as before.

B.3 Multiple evidence arrivals

In this section, we show that Proposition 3 extends to an environment with unlimited evidence arrivals. Suppose each piece of evidence continues to arrive at rate μ , and for simplicity, assume $\lambda_0 = 0$. The definitions of strategy, sequential rationality, belief system, and equilibrium can be extended to this setting.

In what follows, we show that there exists $\bar{p} < 1$ such that if $p_0 > \bar{p}$, there is no equilibrium in which good evidence is always disclosed immediately. Due to $\lambda_0 = 0$, bad evidence is not

disclosed.

Suppose there exists such an equilibrium. When the sender delays (going off the equilibrium path), his private belief diverges from the receiver's belief. Let $F(\pi, \tilde{\pi})$ denote the sender's value function immediately after disclosure of the first piece of good evidence when the receiver's belief is π , the sender's is $\tilde{\pi}$, and the sender does not possess any remaining evidence.

By disclosing at time 0, the sender's expected payoff is simply

$$F(1, 1) = \mathbb{E}\left[\int_0^\infty e^{-rt} \pi_t dt \mid \theta_0 = 1\right] = \mathbb{E}\left[\int_0^\infty e^{-rt} \theta_t dt \mid \theta_0 = 1\right] = \frac{1}{\lambda_1 + r} \quad (95)$$

by the law of iterated expectations.

If the sender discloses at time ϵ , and the sender has no other evidence, the receiver believes the evidence is fresh, but the sender's private belief is just $e^{-\lambda_1 \epsilon}$ accounting for the possibility of unobserved switching since time 0. To a first order approximation, the expected payoff of delaying until time ϵ (and disclosing immediately any good evidence that arrives after time ϵ) is at least

$$p_0 \epsilon + (1 - r \epsilon) F(1, e^{-\lambda_1 \epsilon}). \quad (96)$$

A useful identity is that for each $\hat{\pi} \in [0, 1]$

$$F(1, \hat{\pi}) = \hat{\pi} F(1, 1) + (1 - \hat{\pi}) F(1, 0).$$

This follows from the law of iterated expectations: the sender's expectation of future receiver beliefs equal the sender's expectation, averaging over the current state, of those beliefs conditional on the current state being good or bad. In particular,

$$F(1, e^{-\lambda_1 \epsilon}) = e^{-\lambda_1 \epsilon} F(1, 1) + (1 - e^{-\lambda_1 \epsilon}) F(1, 0). \quad (97)$$

Hence, (96) equals

$$p_0 \epsilon + (1 - r \epsilon) [e^{-\lambda_1 \epsilon} F(1, 1) + (1 - e^{-\lambda_1 \epsilon}) F(1, 0)]. \quad (98)$$

To a first order approximation, the payoff gain from delay is at least

$$\epsilon [p_0 - (\lambda_1 + r) F(1, 1) + \lambda_1 F(1, 0)],$$

obtained by subtracting (95) from (98). Delay is therefore optimal if

$$p_0 > \bar{p} := (\lambda_1 + r)F(1, 1) - \lambda_1 F(1, 0).$$

Since $F(1, 1) = \frac{1}{\lambda_1 + r}$ and $F(1, 0)$ is positive, $\bar{p} < 1$. Note that in our baseline model with one piece of evidence, the receiver's belief after disclosure depends only on the Markov switching dynamics and is therefore independent of the sender's private belief; in this case, $F(1, 0) = F(1, 1)$, so the equation reduces to $p_0 > rF(1, 1) = \frac{r}{\lambda_1 + r}$, reproducing the cutoff in Proposition 3.